

# ANTIHUNTER
# DIGI NODE

Operator's Guide: from the box to the field

Covers all four tiers: **Fully Assembled Unit** · **Complete Parts Kit** · **Populated PCB** · **Bare PCB**

Guide revision 1.1 · 4 October 2026 · Firmware v1.0.5 · Written for the current Stable and Beta firmware channels
github.com/lukeswitz/AntiHunter · Web flasher: lukeswitz.github.io/AntiHunter

# Hello, operator

Welcome to AntiHunter. This node is a self-contained Wi-Fi, Bluetooth and LoRa mesh sensor. It runs everything on the device: no cloud, no account, no server needed. It is fully open source. Read the firmware, write your own mesh commands, wire it into your own IDS. Make it yours.

This guide takes you from opening the box to a tuned node in the field, whichever tier you bought. Sections that only apply to one tier are marked. Everything else applies to every node.

> **READ BEFORE APPLYING POWER: THREE RULES**
>
> **1. Regulated 5 V only.** Power the PCB from USB, a bench supply set to 5 V, a proper DC-DC regulator, or the UPS board. Never connect 18650 cells, a battery pack, or a vehicle supply directly to the board. Unregulated input permanently damages the modules. Voltages under 4.5 V cause instability.
>
> **2. Never power the LoRa radio without its antenna.** Transmitting into an open port can destroy the radio's PA. Fit the LoRa antenna, or the U.FL pigtail plus antenna, before the node is energised.
>
> **3. One USB cable at a time.** The ESP32 and the Meshtastic radio share the PCB's power rails. With both USB ports plugged in, one board back-feeds the other. Plug in one board, do your work, unplug it, then plug in the other.

## What you are holding

| Tier | What is in the box | What you still have to do |
|---|---|---|
| **1 · Fully Assembled Unit** | Built, soldered, tested and ready to deploy. 8 GB SD card fitted. Active GPS helix antenna. SMA antennas. Meshtastic flashed on the radio with the serial link configured. | Fit two 18650 cells, screw on the antennas (§2), flash AntiHunter (§5), set the radio region and channel (§7). |
| **2 · Complete Parts Kit** | Every part on the bill of materials, unassembled, as it comes from the supplier. Passive GPS antenna. Nothing is soldered and nothing is flashed. | Solder and assemble it (§3), flash Meshtastic on the radio (§7.1), then flash AntiHunter (§5) and configure. |
| **3 · Populated PCB** | A fully assembled PCB. Core modules soldered, bench-tested, 8 GB SD card fitted. Factory U.FL antennas. Meshtastic flashed on the radio with the serial link configured. | Attach the antennas (§2), supply power over USB or 18650s, build the housing if you want one, flash AntiHunter (§5), set the radio region and channel (§7). |
| **4 · Bare PCB** | One 82 mm board, unpopulated. | Source the BOM, solder the modules, flash Meshtastic on the radio, fit an SD card, then follow this guide from §3. |

> **TIERS 1 AND 3: THE RADIO IS ALREADY DONE**
>
> On the Fully Assembled Unit and the Populated PCB, the Meshtastic radio ships flashed with the latest stable firmware and the serial link to the node already configured: TEXTMSG mode, 115200 baud, on the correct pins for the board. You do *not* need to redo that. You *do* still need to set your LoRa region and create your own encrypted channel. See §7.
>
> On the Complete Parts Kit and the Bare PCB you flash the radio yourself first. §7.1 has the settings. The ESP32 ships empty on *every* tier, so you always flash AntiHunter yourself and know exactly what is on it.

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# 1 Unboxing and inventory

Open the box on a clean, static-safe surface. Check what arrived against your tier before you power anything.

### 1.1 Tier checklists

**TIER 1: FULLY ASSEMBLED UNIT**

- ☐ Node built, soldered and bench-tested
- ☐ 8 GB SD card fitted
- ☐ Active GPS helix antenna
- ☐ 2.4 GHz Wi-Fi/BLE SMA antenna
- ☐ LoRa SMA antenna for your region
- ☐ Meshtastic flashed on the radio, serial link configured

**TIER 2: COMPLETE PARTS KIT**

Every part needed to build a finished node. Nothing is soldered and nothing is flashed. The two 18650 cells are the only thing you supply.

- ☐ Bare 82 mm PCB
- ☐ XIAO ESP32-S3
- ☐ Heltec V3.2 LoRa radio (or T114)
- ☐ ATGM336H GPS module
- ☐ DS3231 RTC module
- ☐ SW-420 vibration sensor
- ☐ Micro SD reader module
- ☐ 8 GB SD card
- ☐ Pin headers for every module
- ☐ 3D-printed enclosure: main PCB housing, battery housing, front cover, back cover
- ☐ TPU seals for the housing, GPS and USB-C port
- ☐ 3× U.FL-to-SMA pigtail cables

- ☐ SMA bulkheads
- ☐ 2.4 GHz Wi-Fi/BLE antenna, 6 dBi
- ☐ LoRa antenna for your region
- ☐ Passive ceramic GPS antenna, the one supplied with the GPS module
- ☐ 30 mm 5 V fan with JST lead
- ☐ KSD9700 normally-open thermal switch, 40 °C
- ☐ 3-pin mini on/off switch with rubber cover
- ☐ Waterproof Type-C female chassis jack, 2-pin 22 AWG leads, panel nut and dust cap
- ☐ Type-C 15 W 3 A 5 V UPS charger board, dual 18650
- ☐ 6× JST 2.54 2-pin terminals
- ☐ 3× JST power male cables for the switch, thermal switch and power board
- ☐ M3 heat-set inserts, plus 4× M2 for the power board
- ☐ M3×4-6mm screws for the PCB and enclosure lids, M2×4-6mm for the power board
- ☐ 2× M2.5 screws, supplied with the fan
- ☐ 2× 15 mm brass standoffs
- ☐ ¼"-20 tripod insert

**TIER 3: POPULATED PCB**

A fully assembled PCB, bench-tested, on its factory antennas. No enclosure and no power source.

- ☐ PCB with XIAO ESP32-S3, Heltec V3.2 LoRa radio (or T114), ATGM336H GPS, DS3231 RTC, SW-420 vibration sensor and SD reader soldered
- ☐ 8 GB SD card fitted
- ☐ Factory U.FL antennas: flat 2.4 GHz film, coiled LoRa whip, ceramic GPS patch
- ☐ Meshtastic flashed on the radio, serial link configured

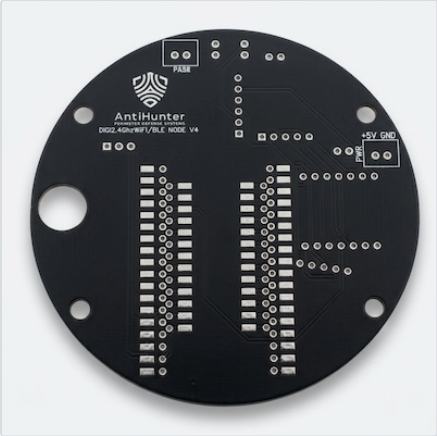
**TIER 4: BARE PCB**

- ☐ One 82 mm AntiHunter DIGI NODE PCB, unpopulated

Everything else comes from the bill of materials in Appendix B, including the Meshtastic radio, which you flash yourself.

## 1.2 What you supply

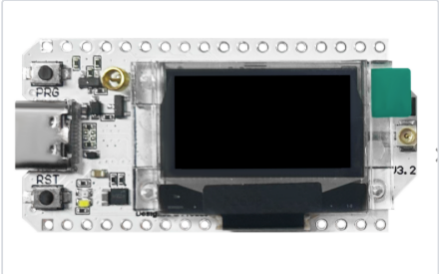
- **Power.** Tiers 1 and 2 take two 18650 cells in the UPS board. Tier 3 runs from USB or from your own regulated 5 V supply. Cells are never supplied on any tier.
- **18650 cells.** Brand new, same brand, same capacity, never mixed old with new. Sourcing, charging, storage and disposal are your responsibility.
- **A micro SD card**, only on the Bare PCB tier. Every other tier ships with an 8 GB card. A replacement must be FAT32, and 32 GB or larger is not recommended.
- **A USB-C data cable.** Charge-only cables will not flash the board.
- **A computer running Chrome or Edge** for the web flasher, or a terminal for the CLI installer.
- **Soldering tools** on tiers 2 and 4: temperature-controlled iron, fine solder, flux, tweezers, small Phillips and hex drivers, heat-resistant surface.



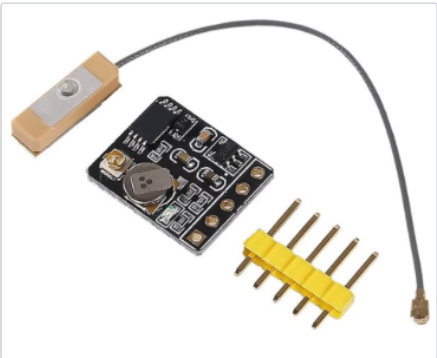

The bare 82 mm board.



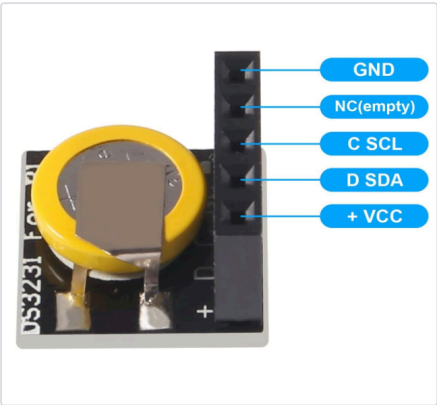

XIAO ESP32-S3.



Heltec V3.2 LoRa radio.



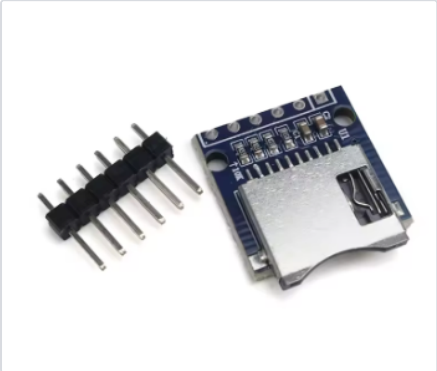
ATGM336H GPS module.




DS3231 real-time clock.



SW-420 vibration sensor. Note the trimpot.



Micro SD reader module.



U.FL-to-SMA pigtail.



5 V UPS charger board.

## 1.3 Power

The node draws roughly 160 mA idle and 600 mA while actively scanning. Inside the enclosure the UPS board provides the regulated 5 V and the cells are never connected to the PCB directly. Outside the enclosure, use a USB-C adapter, a bench supply set to 5.0 V on the 5 V input, or a proper DC-DC regulator.

> **POWER RULES**
>
> Do not connect batteries directly. Do not exceed 5 V. Do not use unregulated sources. A fuse in the battery line is recommended, a 2 A fast-blow in series with the switch wiring, but one is not supplied with any tier.

### 1.4 Thermal

The ESP32 runs warm under load and the GPS module gets noticeably hot while it searches for satellites. Both are normal. Neither the fan nor the thermal switch is required for the node to work.

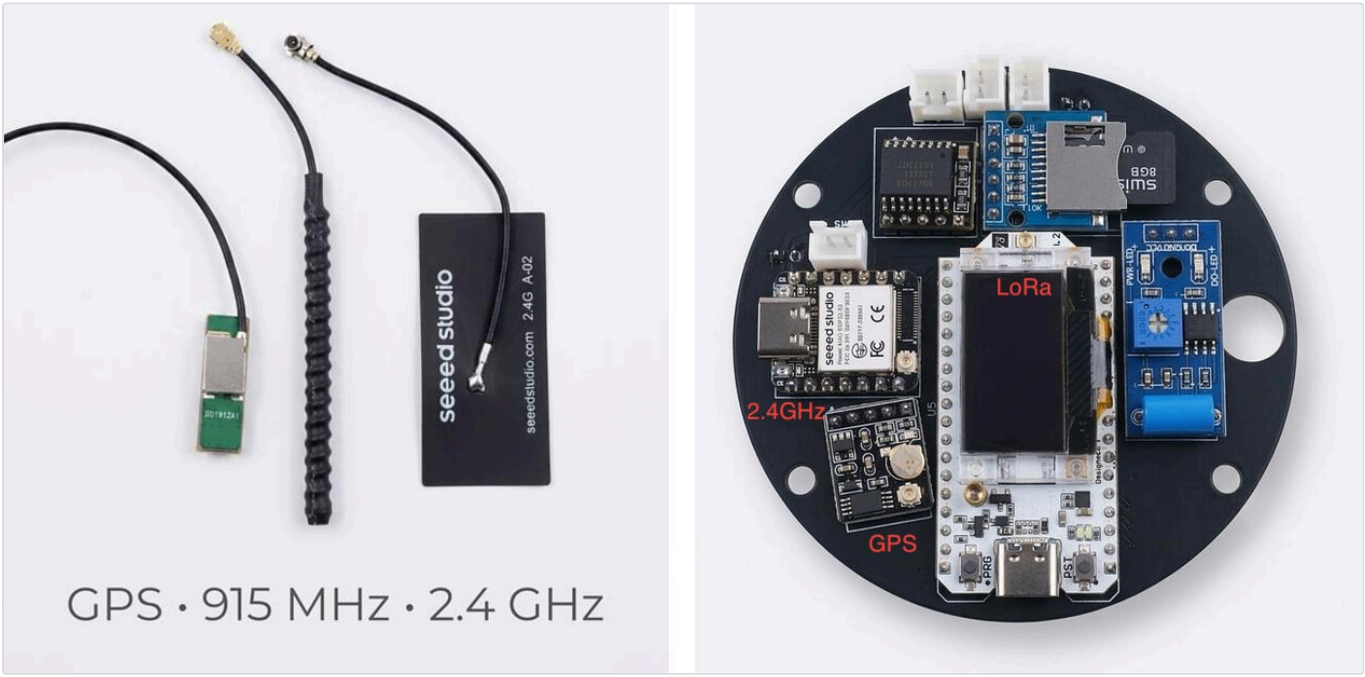
- The 30 mm fan plugs into the FAN JST output.
- The sticker side is not always the exhaust side. Run the fan for a second and feel which way it blows before you screw it down. The Fully Assembled Unit ships with the fan set to exhaust.
- The KSD9700 normally-open thermal switch, 40 °C, wires to the THERMO JST and switches the fan on when the enclosure heats up.
- To run the fan continuously instead, jump the THERMO input or supply 5 V straight to the fan port.

## 2 Antennas

More nodes are damaged fitting antennas than in any other step. The U.FL connectors are small, fragile and unforgiving of a crooked press. Take your time here.

### 2.1 Which antenna goes where

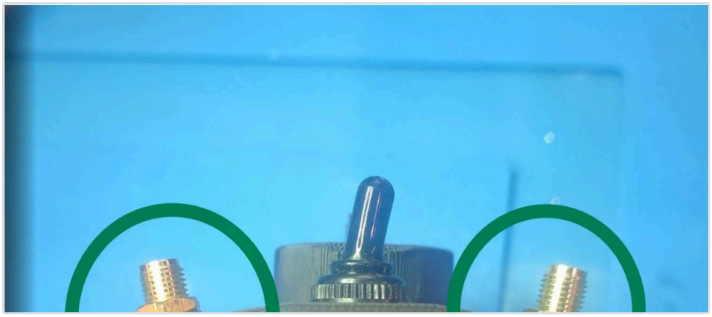
| Antenna | U.FL set (tiers 2 and 3) | SMA set (tiers 1 and 2) | Connects to |
|---|---|---|---|
| **2.4 GHz Wi-Fi / BLE** | Flat black film antenna, printed 2.4G | Shorter of the two rubber ducks, 6 dBi | The U.FL socket on the XIAO ESP32-S3, or its SMA bulkhead |
| **LoRa** (region specific) | Coiled black whip | Longer duck, marked 868 MHz (EU), 915 MHz (US) or 923 MHz (Asia) | The U.FL socket on the Heltec / T114 radio, or its SMA bulkhead |
| **GPS** | Passive ceramic patch, supplied with the GPS module | Active helix, Fully Assembled Unit only | The U.FL socket on the GPS module, or the SMA bulkhead in the battery housing |




The factory U.FL antenna set beside the sockets each one belongs to: ceramic GPS patch, coiled LoRa whip (915 MHz shown, yours matches your region), flat 2.4 GHz film antenna. The GPS module carries the GPS socket, the Heltec carries LoRa, the XIAO carries 2.4 GHz Wi-Fi and BLE.

**PICK AN ORIENTATION AND KEEP IT**

On an enclosed build, decide which bulkhead carries LoRa and which carries Wi-Fi, then wire every node you build the same way. It does not matter which you choose. It matters that a spare antenna always fits the port you expect in the field.



The two SMA bulkheads with the power switch centered between them.

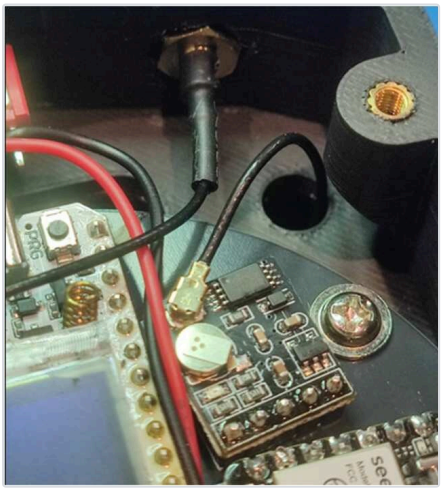
**MATCH YOUR ANTENNAS ACROSS NODES**

When you build more than one node, use the same antenna type and the same cable length on all of them. Triangulation and any other multi-node distance estimate assumes the nodes hear the world the same way. Mismatched antennas skew the result.

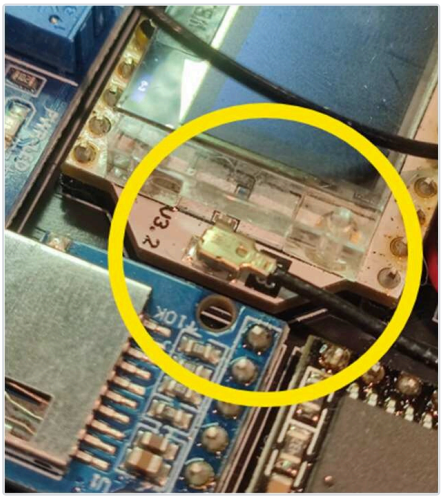
## 2.2 Fitting a U.FL / IPX pigtail (tiers 2 and 3)

The pigtail's tiny snap connector mates with the socket on each module. It seats with an audible, tactile click.

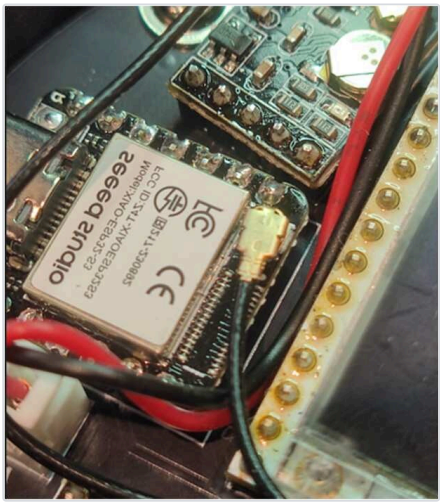
1. Identify each pigtail and the socket it belongs to. Route the cable before you connect it, not after.
2. Hold the connector **parallel to the board** and align its rim concentrically over the socket. Look at it from the side: the two rims must be perfectly stacked.
3. Press **straight down** with a fingertip or the flat of a plastic tool until it clicks.
4. Tug the cable gently to confirm it is home. If it lifts off, it never seated.



GPS module, connector facing up.



LoRa radio, socket circled.



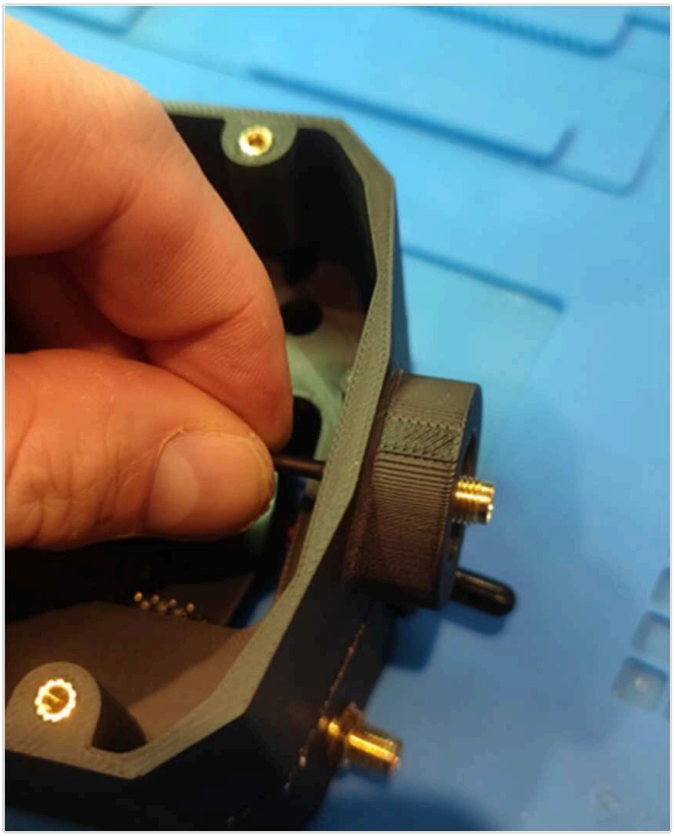
XIAO ESP32-S3, Wi-Fi/BLE socket.

**U.FL: DO NOT**

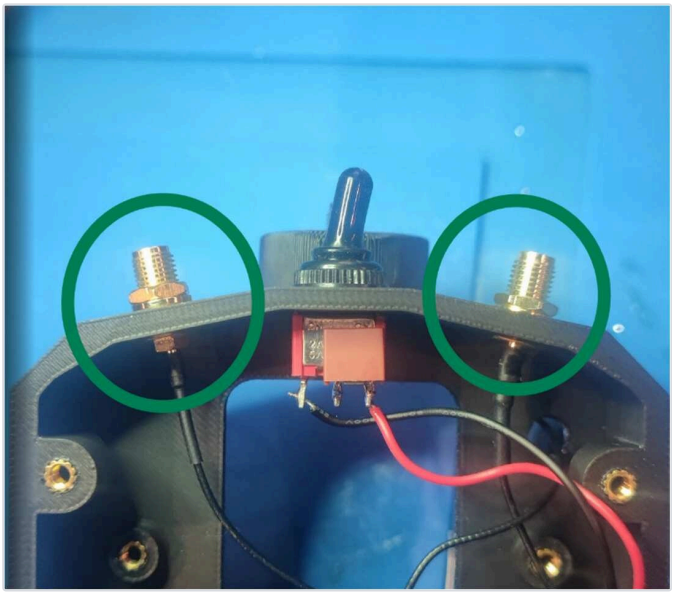
- Do not press at an angle.
- Do not force a connector that will not click.
- Do not pull the cable to remove one. Lift the connector body straight up, or use a U.FL removal tool.
- Do not reseat one repeatedly. The contacts wear out.
- **Misalignment damage is permanent.** A crushed socket means a new module.

### 2.3 Fitting SMA antennas (tiers 1 and 2)

1. Check the bulkhead is tight in the shell. The nut and washer go on the outside, and the bulkhead should not rotate when you turn an antenna onto it.
2. Start the thread by hand, straight on. Never cross-thread it.
3. Turn until finger-tight, then no more than an eighth of a turn. Do not use pliers.
4. Fold the duck upright once fitted. Antennas work best vertical.



GPS SMA female fitted into the battery housing, secured from outside with washer and nut.



Wi-Fi SMA (female, left) and LoRa SMA (male, right) in the main housing.

**WET CONDITIONS**

A smear of silicone grease on the SMA threads and on the TPU seals keeps water out and makes the antennas easier to remove later.

## 3 Assembly (tiers 2 and 4)

Tiers 1 and 3 arrive soldered. Skip to §4.

> **BUILD FROM THE ASSEMBLY MANUAL, NOT FROM HERE**
>
> The Complete Parts Kit and the Bare PCB are built with the illustrated assembly manual, which covers soldering, connectors, the enclosure, the SD card and the power wiring in photographed steps. This guide does not repeat it.
>
> `hw/Prototype_STL_Files/Antihunter-DIGINODE-AssemblyManual.pdf`
> `github.com/lukeswitz/AntiHunter`, in the `hw` folder
>
> Work through it end to end, then come back here at §4 to power the node up for the first time.

# 4 First power-on

### 4.1 What a healthy boot looks like

Connect a USB-C cable to the **ESP32** only, remembering one USB at a time, and open a serial monitor at 115200 baud. On boot the firmware prints its banner and version, then the results of its hardware checks: SD card, GPS, RTC, vibration sensor. Any module that failed to initialize says so by name.

If you flashed the **Full** firmware, a Wi-Fi access point named `Antihunter` also appears within a few seconds.

### 4.2 Confirming the hardware

Every subsystem reports into the diagnostics text, visible in the web UI's System tab, over serial, and in the reply to a mesh `STATUS` command. Check for:

- **SD card.** Mounted and writable. If it is missing, the node still scans but nothing is logged.
- **GPS.** Reports a fix once it has one. Give it a clear view of the sky; indoors it may never lock. Detections carry coordinates only when the fix is valid.
- **RTC.** Supplies the timestamps on every log line and export.
- **Vibration sensor.** Shows `Standby`, or `Active` with the time of the last movement.

### 4.3 The clock

Nothing to do. The DS3231 is set when you flash the firmware, and GPS disciplines it again on every fix. Its coin cell keeps it running across power cycles. The clock keeps UTC. Logs, capture file names, results and heartbeats show local time, with daylight saving, once a GPS fix places the node in a known zone: Hawaii, Alaska, Arizona, Saskatchewan, US/Canada Pacific, Mountain, Central and Eastern, UK/Ireland, Central Europe, Eastern Europe. Elsewhere, and until the first fix after boot, they show UTC.

> **ORDER OF OPERATIONS**
>
> Flash the firmware first (§5), then set the clock and configure the radio. A fresh flash with "Erase Device" ticked clears saved settings, so do it before you spend time configuring.

# 5 Flashing the firmware

The ESP32 ships empty on every tier. You choose the build and put it on yourself.

### 5.1 Full or Headless?

Same firmware, same detectors, same mesh commands, same scan engine. The difference is whether the node hosts a Wi-Fi access point.

| | Full | Headless |
|---|---|---|
| **Control** | Web UI and HTTP API over its own AP, plus serial and mesh | Serial and mesh only |
| **RF footprint** | The AP beacons continuously, so the node is discoverable as a Wi-Fi network | Never beacons |
| **Results** | Rendered as cards in the web UI's Results tab, and served plain over the API like every other endpoint. Written to the SD card as well | Written to `/last_results.txt` on the SD card |
| **Logging** | Identical. Both builds write results, the probe database, vibration events, sentinel incidents and scan logs to the SD card | |
| **Device database** | `/devicedb.jsonl`, which is what the Data Explorer browses | Not written |
| **Best for** | Bench work, first setup, anything where you want the UI | Covert and long-term field deployments |

Start with Full while you learn the device. Reflash to Headless before a quiet deployment. It takes two minutes and your settings live on the SD card.

### 5.2 Release channels

| Channel | Source | Use it when |
|---|---|---|
| **Stable** | `main` branch | Default. Field deployments. |
| **Beta** | `beta` branch | New features first, with the risk that implies. |
| **Experimental** | ESP32-C5 builds (2.4 + 5 GHz) and RadarNode | Only on C5 hardware. The flasher asks you to acknowledge these are test builds. |

### 5.3 Method A: web flasher (recommended)

1. Open **lukeswitz.github.io/AntiHunter** in Chrome or Edge on a desktop. Other browsers cannot talk to serial ports.
2. Choose **Full** or **Headless**, and a release channel.
3. Plug the ESP32 in with a data cable, nothing else connected to the board.
4. Click **Connect & Flash** and pick the port.
5. Optionally set your device parameters before flashing. The **Sentinel & Detectors** panel configures the whole detection engine: start-on-boot, radio mode, every detector toggle, mesh flags, thresholds. Anything left on *Default* keeps the firmware's own setting.

> **ERASE DEVICE**
>
> Tick it when upgrading from firmware older than v0.9.2, or when you want to clear saved settings. Otherwise leave it off. Settings persist in NVS and are mirrored to the SD card, and self-heal if they are corrupted.

### 5.4 Method B: command line

```
curl -fsSL -o flashAntihunter.sh \
  https://raw.githubusercontent.com/lukeswitz/AntiHunter/main/Dist/flashAntihunter.sh
chmod +x flashAntihunter.sh
./flashAntihunter.sh
```

The script asks for a release channel, then Full or Headless. Flags: `-c` configure device parameters during the flash, `-e` erase flash first, `-l` list available firmware.

### 5.5 Method C: build from source

```
git clone https://github.com/lukeswitz/AntiHunter.git
cd AntiHunter
pio device list                                # find the port
pio run -e AntiHunter-full -t upload           # web UI build
pio run -e AntiHunter-headless -t upload       # serial + mesh build
pio device monitor -e AntiHunter-full          # watch it boot
pio run -e AntiHunter-full -t erase -t upload  # clean flash
```

Environments: `AntiHunter-full` and `AntiHunter-headless` for the ESP32-S3; `AntiHunter-c5-full` and `AntiHunter-c5-headless` for the ESP32-C5 on the `feat/c5` branch; `RadarNode-c5` for the 24 GHz radar node.

### 5.6 Verifying the flash

- The serial monitor prints the banner and firmware version at boot.
- Full build: the `Antihunter` access point appears.
- Connect to it, password `antihunt3r123`, and open **http://192.168.4.1**. Note `http`, not https.

> **DO THIS BEFORE ANYTHING ELSE**
>
> The default AP name and password are published in the documentation. Until you change them in **RF Settings**, treat the node as an open door. §8.1.

### 5.7 Board variants

The XIAO ESP32-C5 is a drop-in replacement for the S3 on the same board, with the same pads and peripherals, and it adds 5 GHz scanning. Those builds are in testing and live under the flasher's Experimental channel. The RadarNode variant pairs a 24 GHz radar with the same mesh: it detects a moving target, then sweeps Wi-Fi and BLE to record which devices were present at that moment.

## 6 Staying up to date

### 6.1 Where updates come from

Firmware is built and published by CI when a release is tagged. The web flasher always serves the current build for the channel you pick, so re-running the flasher is the update. Nothing on the node phones home or updates itself.

### 6.2 Getting notified

1. Open **github.com/lukeswitz/AntiHunter** and sign in.

2. Click **Watch** → **Custom** → tick **Releases** → Apply. You now get an email or notification for every release, without the noise of every commit.
3. Read the release notes before flashing: `docs/release-notes/notes-stable.md` and `notes-beta.md` in the repository, and on each release page.

Discussions and issues on the same repository are where behavior changes, hardware alternatives and field reports get talked through.

### 6.3 Updating a deployed node

1. Bring it in, or reach it over USB in place.
2. Re-run the web flasher or the CLI script on the same channel. Leave **Erase Device** unticked to keep your settings.
3. Reconnect and check the version, then confirm your targets, node ID and detector settings survived.

Settings live in NVS and are mirrored to the SD card, so they normally survive an update. Watchlists, results and databases live on the SD card and are untouched by flashing.

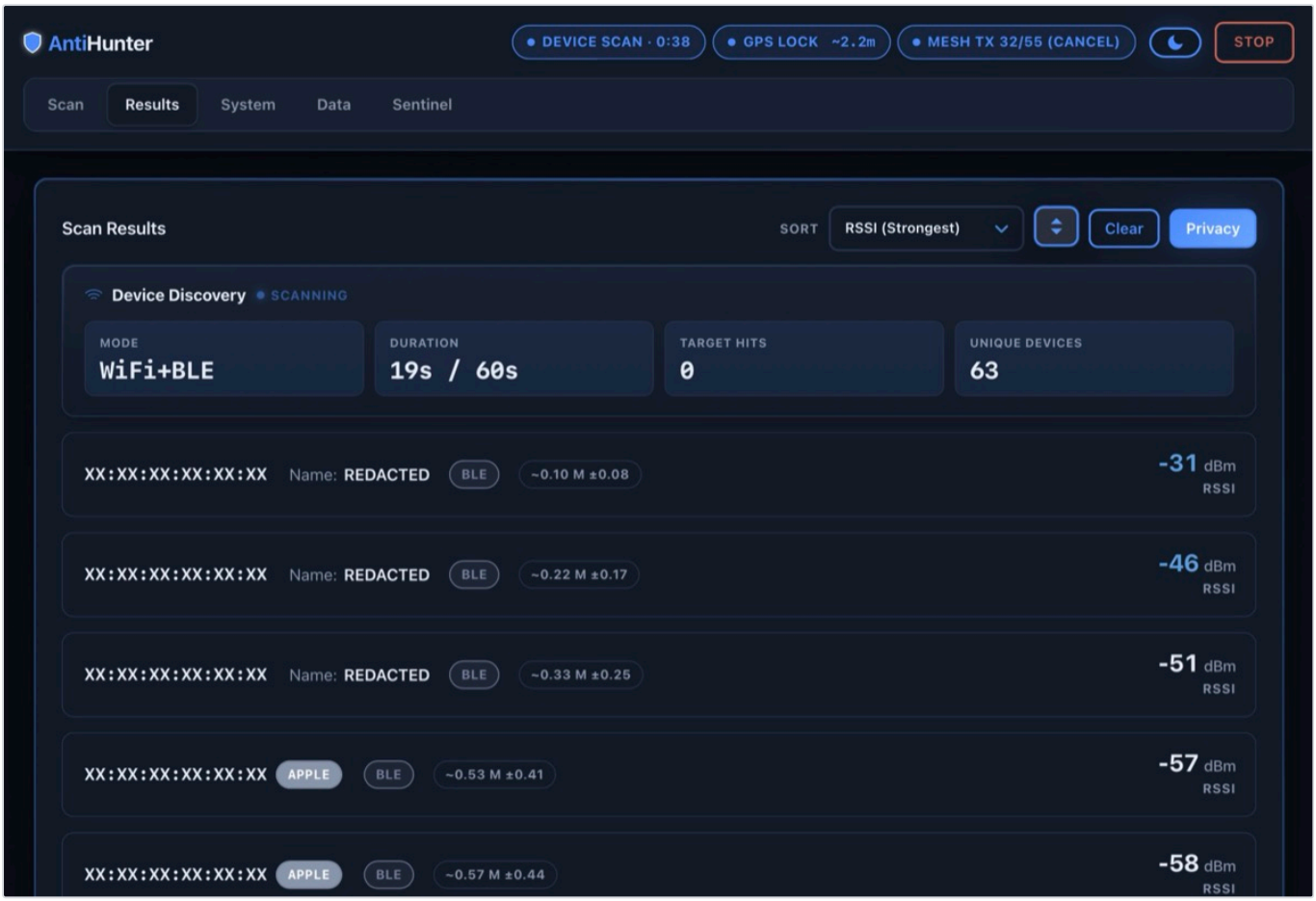
### 6.4 Rolling back

Flash the other channel, or check out an earlier tag and build from source with PlatformIO. If a downgrade behaves strangely, erase and flash clean, then reconfigure.

> **FLEET DISCIPLINE**
>
> Run one firmware version across a mesh. Mixed versions can differ in the mesh message formats they emit and what your Command Center is able to parse.

## 7 Mesh radio setup

The LoRa mesh gives you long-range control and alerting with no infrastructure. It is entirely optional, since a single node works standalone, but it is what turns several nodes into a system.




A device scan sending its results over the mesh. The header badge tracks the transmit queue draining at the LoRa airtime cap, 32 of 55 rows sent, and Privacy Mode has blanked the addresses.

### 7.1 Flashing and linking the radio

**Tiers 1 and 3: already done.** On the Fully Assembled Unit and the Populated PCB the radio arrives on the latest stable Meshtastic with the serial link to the node set: **TEXTMSG** mode, **115200** baud, on the right pins for the board (19 RX / 20 TX on Heltec V3, 10 RX / 9 TX on T114). The screen is set to blank after 1 s and the status LED is off. Verify it if you like; do not redo it.

**Tiers 2 and 4: you flash it.** The Complete Parts Kit and the Bare PCB include no flashed firmware at all. Connect the Heltec V3.2 (preferred) or T114 on its own USB and flash the latest stable Meshtastic from **flasher.meshtastic.org**. Then set the serial module, either in the Meshtastic app or web client, or with the script in §7.2:

| Setting | Value |
|---|---|
| Serial module | Enabled |
| Mode | `TEXTMSG` |
| Baud | `115200` |
| Pins, Heltec V3 | 19 RX / 20 TX |
| Pins, T114 | 10 RX / 9 TX |

Without this the node and the radio cannot talk, and no mesh command will ever reach the node.

> **THREE THINGS EVERY RADIO NEEDS, WHATEVER TIER YOU BOUGHT**
>
> - **LoRa region.** Defaults to `UNSET`. The radio receives but will not transmit until you set it (§7.2).
> - **Bluetooth pairing PIN.** Every unit has the same default pin. Change it, or turn BLE off.
> - **Channel.** Defaults to the public channel. Make your own encrypted channel primary and turn the public one off (§7.3).

### 7.2 What you must set: region

The LoRa region is not set at the factory, and an `UNSET` region makes the radio receive-only: it hears the mesh but never transmits. Set the region for the country you are in. It determines the frequency band and the legal power and duty-cycle limits.

Plug the **radio** in on its own USB, not the ESP32, and run:

```
pip3 install 'meshtastic[cli]' pyserial
python3 scripts/meshtastic_config.py
```

With no arguments it prints the current settings and opens an interactive menu. Non-interactive equivalent:

```
python3 scripts/meshtastic_config.py --serial on --board heltec-v3 \
    --screen off --led off --ble off --region US
```

| Option | What it does |
|---|---|
| `--port` | Serial port, auto-detected if omitted |
| `--board heltec-v3 \| t114` | Sets the serial-module pins for your board |
| `--region` | LoRa region, e.g. `US`. `UNSET` means receive only |
| `--serial on \| off` | The node link: TEXTMSG at 115200 on the board's pins |
| `--screen on \| off \| SECS` | `off` blanks after 1 s, or give a timeout in seconds |
| `--led on \| off` | Status LED heartbeat |
| `--ble on \| off` | Bluetooth radio |
| `--pin NNNNNN \| none \| random` | BLE pairing: fixed 6-digit PIN, none, or random |

The same settings can be applied from the Meshtastic phone app or web client if you prefer.

### 7.3 Your own encrypted channel

Meshtastic defaults to a public channel whose key is published. Anything you send on it, commands, detections and coordinates alike, is readable by anyone in radio range running stock firmware. Before you deploy:

1. In the Meshtastic app, create a **new channel** with a generated key and give it a name of your own.
2. Make it the **primary** channel.
3. **Turn the public default channel off.**
4. Share the channel to your other nodes and to your phone by QR code or channel URL. Every node in your mesh must carry the same channel.

> **THE FACTORY PRESET IS NOT PRIVATE**
>
> A node left on the default channel broadcasts its detections, its position and its command traffic to anyone listening. Changing the channel is the single most important mesh setting you will touch.

### 7.4 Quiet-ops settings

- **Screen off** and **LED off** removes the visual signature and saves power. Leave the screen in place and run `python3 scripts/meshtastic_config.py --screen off --led off`.
- **Assembly:** the Heltec screen does not need to come off. Unscrew the three screws under it, hold the screen back to reach the pins near the glass and flex cable, solder them, then put the screen back and refit the screws.
- **Screenless node (optional):** the screen can be removed permanently, but this is advanced and at your own risk. Heltec states the display is soldered on and publishes no removal procedure, so there are no vendor steps to follow. Without a screen, configure the node over serial: set the pairing PIN with `python3 scripts/meshtastic_config.py --pin NNNNNN` (6 digits, or `none`), since `random` would show the PIN on the now-absent screen. The same script, or the Meshtastic serial CLI, configures everything else.
- **BLE off** on the radio removes its Bluetooth signature. Set a pairing PIN first if you intend to turn it back on later.
- **Headless firmware** on the node gives no Wi-Fi AP at all.

### 7.5 Addressing and node IDs

- Node IDs are 2 to 5 alphanumeric characters, A-Z and 0-9, no spaces. Give every node a distinct one.
- `@ALL COMMAND` broadcasts to every node. `@AH01 COMMAND` targets one.
- Outbound messages are rate-limited to protect airtime: roughly 167 bytes per second sustained, with device rows packed into frames of up to 230 bytes.

### 7.6 Reaching the mesh from a phone, TAK or MQTT

Commands and detections are ordinary Meshtastic text messages, so anything that talks to the mesh talks to your nodes:

- **Phone.** Pair the radio to the Meshtastic app over Bluetooth or Wi-Fi and send `@node COMMAND` messages from anywhere in mesh range. No AP, no server.
- **TAK / ATAK.** The Meshtastic ATAK plugin and TAK server integration bridge the whole mesh, node traffic included.
- **MQTT.** A Meshtastic MQTT gateway node forwards mesh traffic to a broker for logging, Home Assistant or Node-RED.

These are Meshtastic features. Nothing AntiHunter-specific is needed on the receiving end.

## 8 First configuration

Ten minutes on the bench before the node ever leaves the building.

### 8.1 Change the credentials

- **AP SSID and password.** The defaults `Antihunter` and `antihunt3r123` are published. Change both under **RF Settings**.
- **Hidden network.** Off by default. Turn it on under **RF Settings** and the node stops beaconing its SSID, so the AP does not appear in a scan of the area. It is not access control. A hidden SSID forces the client to send directed probe requests carrying the network name, so the phone or laptop you connect with advertises that name

everywhere it goes afterwards. The WPA2/WPA3 passphrase is still what keeps people out. The node reboots on save and you type the SSID by hand to reconnect.

- **Erase key.** Authorizes every wipe, abort, auto-erase change and factory reset. 8 to 64 characters, no `:`. Erase commands are refused (`ERASE_ACK:SET_KEY_FIRST`) until you set one: in the web UI, in the flasher (`erasePSK`), or with `@AH01 CONFIG_ERASE_PSK:<new>:<current>`. Full: until a key is set, `<current>` is the AP password, once it is changed from the default. Headless: set it in the flasher. Record it; the node never prints it.
- **Meshtastic BLE pairing PIN.** Set a fixed PIN or turn BLE off entirely.

### 8.2 Identity

- **Node ID.** Set it in the web UI or with `@AH01 CONFIG_NODEID:AH02`. It prefixes every mesh message the node sends, so make it meaningful.
- **Clock.** See §4.3. Timestamps are local time once GPS fixes the zone, UTC before that.

### 8.3 Scan presets

Presets set the whole timing profile in one click. Start on Balanced.

| Preset | Wi-Fi chan time | Wi-Fi scan int | BLE scan int | BLE scan dur | RSSI | Use case |
|---|---|---|---|---|---|---|
| Relaxed | 300 ms | 5000 ms | 6000 ms | 3000 ms | –80 dBm | Low power |
| Balanced | 160 ms | 3000 ms | 4000 ms | 2000 ms | –95 dBm | General use (default) |
| Aggressive | 110 ms | 1500 ms | 2000 ms | 1000 ms | –100 dBm | Fast detection, high coverage |
| Custom | User-defined | | | | | Fine-tuned |

#### What each parameter does

- **Wi-Fi channel time**, 50 to 300 ms. Dwell per channel. It must clear the ~100 ms beacon interval to catch every AP on a channel. Shorter covers the band faster but risks missing beacons.
- **Wi-Fi scan interval**, 1000 to 10000 ms. How often the all-channel AP discovery scan runs. Between scans the node captures frames passively while hopping.
- **BLE scan interval**, 1000 to 10000 ms. Time between BLE cycles.
- **BLE scan duration**, 1000 to 5000 ms. Active BLE scanning per cycle.
- **RSSI threshold**, –100 to –10 dBm. Global signal filter. Triangulation ignores it.
- **Wi-Fi channels.** Comma-separated, `1,6,11`, or a range, `1..14`. Default is 1 to 11.

> **ONE RADIO, TWO JOBS**
>
> Wi-Fi and BLE share a single 2.4 GHz radio and the scan loop is single-threaded. A BLE scan holds the radio for its full duration, and Wi-Fi capture is off-air for that time. BLE scan duration is literally the fraction of each cycle Wi-Fi is dark. The presets split it evenly by setting BLE duration to half the BLE interval.

### 8.4 Field controls worth knowing now

- **Privacy Mode.** One click blanks MACs, SSIDs and GPS in the web UI for screenshots and demos. **Exports and SD data are not redacted.** Scrub those separately.
- **Allowlist.** A global list of devices to ignore across every scan mode. Put your own kit on it.
- **Battery Saver.** Stops Wi-Fi and BLE scanning, drops the CPU to 80 MHz and sends a heartbeat (temperature, GPS) every interval. Mesh stays up. Start it with `@AH01 BATTERY_SAVER_START:10`.
- **Accent Colors.** In the System tab. Sets the color of the destructive controls — STOP, the wipe buttons, Clear, error toasts and danger badges — of the Sentinel banners and of the movement hit color. The choice is held in the

browser, so a second phone or laptop keeps its own, and the node is not written to.

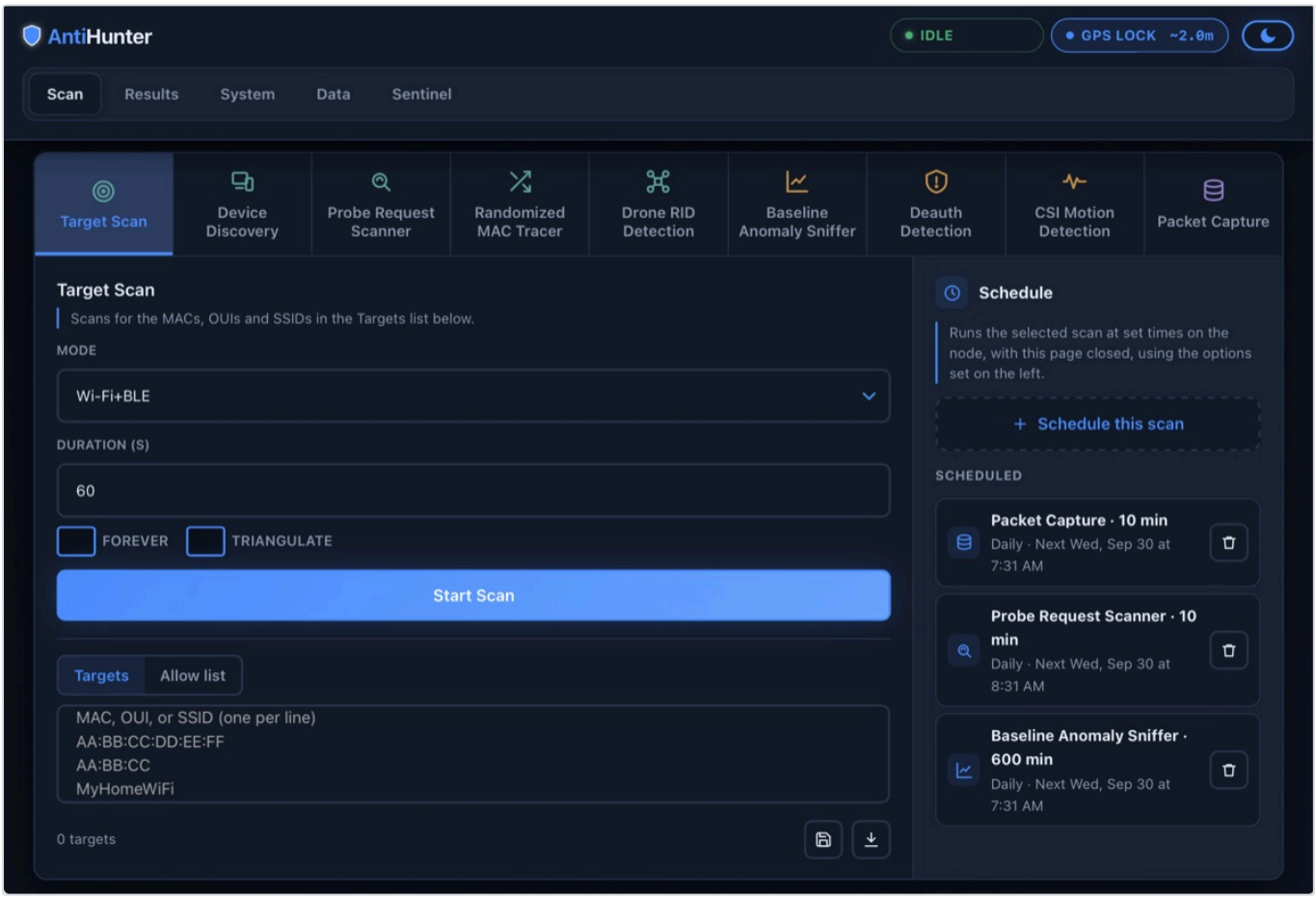
- **Fleet** (System tab, full). Lists the nodes and mesh radios heard on your channel. **Ping Nodes** sends `@ALL STATUS` so every node answers; **Clear** empties the list.
- **Mesh dedup TTL.** Suppresses re-broadcasting a MAC already sent within the window, default 300 s, max 3600. Saves airtime on repeated scans of the same environment. Never applied to alerts, drone events or triangulation.

### 8.5 Where settings live

Configuration is written to NVS on the ESP32 and mirrored to the SD card, so it survives reboots and normal reflashes, and self-heals if a copy is corrupted. Ticking **Erase Device** at flash time wipes NVS deliberately.

## 9 Using the node

Every capability runs on the node itself and reports to the web UI, the mesh and the Command Center alike. One scan mode runs at a time.




The Scan tab on the Full firmware: pick a scan in the top row, set its options on the left, schedule it on the right. Every scan also runs from a mesh command.

### 9.1 Target Scan

A watchlist scan. Load MAC addresses, OUI prefixes, the first three bytes of a MAC so a whole vendor, or SSIDs, and get an instant mesh alert when one is seen.

```
@ALL CONFIG_TARGETS:AA:BB:CC:DD:EE:FF|11:22:33|MyNetwork
@ALL SCAN_START:2:300:1..11     # mode 0=Wi-Fi 1=BLE 2=both : seconds : channels
```

Alert: `NODE: Target: MAC RSSI:dBm Type:WiFi|BLE [Name:name] [GPS=lat,lon] [V=vendor]`. `V=` is the first word of the vendor name, sent when the MAC is not randomized.

In the web UI a target card shows **Getting closer** or **Moving away** when its signal has clearly risen or fallen over recent sightings. Nothing shows while the signal holds steady, or before the target has been seen four times.

### 9.2 Device Scanner

Captures everything nearby, Wi-Fi and BLE, with RSSI, channel and names. This is your baseline survey tool. Run it first at any new site to see what normal looks like.

```
@ALL DEVICE_SCAN_START:2:300:+PROBE   # +PROBE also fills the probe database
```

**What a BLE device is** (full firmware). Device, Target and Baseline results show a badge naming what the device advertises: an Apple service (AirDrop, Handoff, AirPods, Find My, iPhone activity state), or a class and vendor read from its Bluetooth service or company ID, for example `Wearable–Garmin`, `Tag–Tile`, `Vehicle–Tesla`, `Drone–RemoteID`. The text results carry it as `C=`. Wi-Fi devices with a fixed MAC show their vendor from the MAC prefix instead.

### 9.3 Probe Request Scanner

Passive sniffing of the probe requests devices emit while looking for networks they know. It reveals the SSIDs a device is searching for, which is often more identifying than the device itself.

```
@ALL PROBE_START:2:300:+ALL     # +ALL broadcasts every probe, not just watchlist hits
```

Hits: `NODE: PROBE_HIT MAC [Randomized|Vendor] RSSI=dBm CH=N [SSID="net" [GHOST]]`

**Ghost SSID detection** flags a probed SSID with no matching AP responding nearby: a network the device remembers but that is not here.

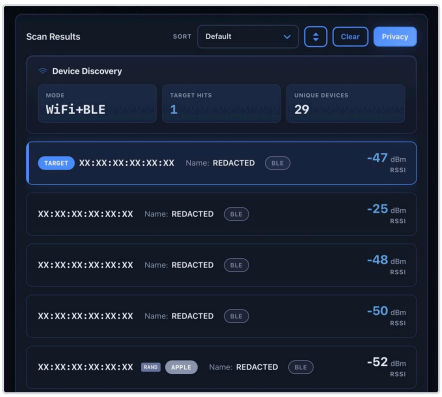
### 9.4 Baseline Anomaly Detection

Learn, then alert. The node observes a site for a set period and records what belongs there. Afterwards it reports what is new, what has gone, and what has changed signal.

```
@ALL BASELINE_START:300        # minimum 60s; add :FOREVER to run indefinitely
@ALL BASELINE_STATUS
```

Alerts: `ANOMALY–NEW`, `ANOMALY–RETURN`, `ANOMALY–RSSI`, `DEVICE_DISAPPEARED`.

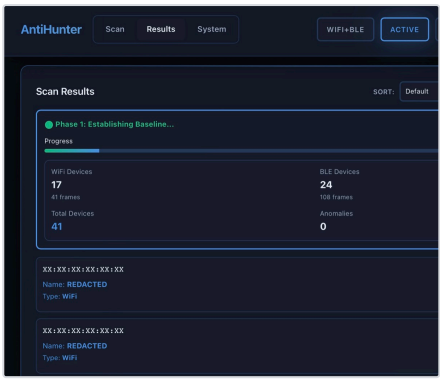
**Watch for changes from now**. While a baseline runs, press the button (full firmware) or send `@ALL BASELINE_WATCH`. The node replies `BASELINE_ACK:WATCHING`, or `BASELINE_ACK:NOT_RUNNING` if no baseline is running. When the baseline ends, it reports what is new here, what is gone, and what moved closer or away since that moment. Full firmware shows this in a *Since* section in Results. Headless prints the device list to serial and sends `BASELINE_WATCH: New=<n> Gone=<n> Moved=<n> Both=<n>` over mesh. Use it when you leave a site and want to see what changed while you were away.




Target scan results (§9.1).




Probe detection: devices, probes, SSIDs and saved records (§9.3).




Baseline learning phase: device counts and the anomaly total (§9.4).

## 9.5 MAC Randomization Correlation

Modern phones rotate their MAC address to defeat tracking. This mode links rotating MACs back to one persistent identity using behavioral signatures, and reports the identity rather than the address.

```
@ALL RANDOMIZATION_START:2:300
```

Emits `IDENTITY:` messages. Treat the output as an investigative lead, not proof. This is an experimental mode.

## 9.6 Deauth Attack Detection

Real-time detection of deauthentication and disassociation frames, with the source tracked.

```
@ALL DEAUTH_START:300
```

Alert: `NODE: ATTACK: DEAUTH|DISASSOC [BROADCAST|TARGETED] SRC:MAC DST:MAC RSSI:dBm CH:N R:reason`. The IEEE reason code is reported as context only: codes 1, 2, 6 and 7 all occur in normal traffic, so the firmware does not use the code to decide whether a frame is an attack.

## 9.7 Sentinel counter-intelligence (beta)

Passive detection of attacker *tooling*, not just attacks: deauth, beacon, auth and association floods, SAE denial of service, karma and evil-twin access points, probe floods, and handshake or PMKID capture attempts. Where a frame carries a recognizable template, Sentinel names the tool behind it.

```
@ALL SENTINEL_ON
@ALL SENTINEL_MODE:scan        # hop the configured channels
@ALL SENTINEL_BOOT:1           # start automatically on boot
@ALL GROUP:dos:on              # enable a detector group
@AH01 SENTINEL_STATUS
@AH01 INCIDENTS:50             # dump the incident log to serial
```

| Group | Detectors |
|---|---|
| `dos` | eviltwin, sae, assoc_sleep |
| `rogue` | eviltwin, owe, karma |
| `recon` | pmkid, probe_flood, hshk |
| `physical` | frag, tsf, jam |
| `mesh` | mesh_guard |

| Group | Detectors |
| --- | --- |
| `all` | every member above |

Deauth, beacon and auth detection are always on and have no toggle. `SENTINEL_MODE:defend` pins one channel. On Headless there is no AP to pin to, so use `scan`. Sentinel is Wi-Fi only. BLE and drone detection are separate modes.

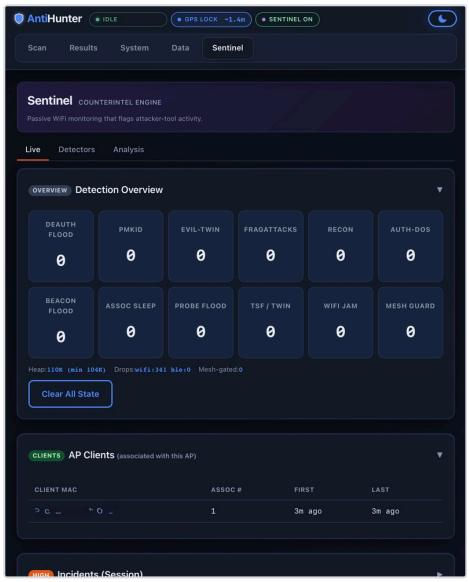
> **KARMA BAIT**
>
> Karma bait is the only Sentinel detector that transmits: a probe request every 8 seconds to bait a karma AP into responding. It is off by default. Enable it deliberately with `GROUP:rogue:on` and only where transmitting is appropriate.

### Acting on a confirmed attack

The **On confirmed attack** panel picks what runs when a detector confirms an attack with a source MAC: triangulate, packet capture, device discovery, probe sweep, drone RID, each with its own duration. Only one can hold the radio, so several ticked run in that order, one at a time, as it frees up. The queue is dropped after 15 minutes if the radio never frees.

Same panel sets the auto-capture budget. Captures made this way are `auto_` files, pruned oldest-first to stay inside the budget and keep the card's free-space floor. Manual captures are never touched.

> **A RESPONSE IS A SCAN**
>
> Detection pauses for the duration of each action. On a site that trips often, keep the durations short.



The Sentinel tab: a live counter per detector, the associated AP clients it has seen, and the session incident log.

## 9.8 Drone Remote ID

Identifies drones broadcasting Remote ID (ODID / ASTM F3411, and French ID) over both Wi-Fi beacon/NAN and BLE. Reports the serial or CAA registration, position, altitude and speed, and the operator's location when the aircraft broadcasts it.

```
@ALL DRONE_START:300
```

`DRONE:` is sent once per appearance. A drone that stays in range is not re-announced. `DRONE_LOST:` follows 120 s after the last beacon.

## 9.9 Triangulation

Several nodes each report RSSI for one target MAC, and the coordinator fits a position from the set. Needs at least three nodes with valid GPS fixes and matched antennas.

```
@AH01 TRIANGULATE_START:AA:BB:CC:DD:EE:FF:60:2:1.0:1.0
@AH01 TRIANGULATE_RESULTS
@ALL TRIANGULATE_STOP
```

The environment argument sets the path-loss model: 0 open sky, 1 suburban, 2 indoor, 3 dense indoor, 4 industrial. Choosing the wrong one is the usual cause of a distance estimate that looks wrong. Results arrive as `T_D:` per-node

data, `T_F:` the final fix with confidence and uncertainty, and `T_C:` complete, with a map link.

### 9.10 CSI Motion Detection (experimental, beta firmware only)

Tells you when someone is moving nearby, even through walls. Indoors only. It notices movement, not someone sitting still.

**How it works.** Wi-Fi signals bounce around a room. When a person moves, the bounces change. The node listens to the Wi-Fi routers and phones around it and alerts when several of their signals change at once.

**Start it:** Scan tab → CSI Motion Detection → Start Scan. Over mesh:

```
@ALL CSI_MOTION_START:0:FOREVER
```

**Pick a sensitivity:**

- **Low** (default): fewest false alarms. Two devices must see the movement.
- **Medium**: more sensitive. Two devices, or one device seeing a very strong change (4× the trigger).
- **High**: most sensitive. Same as Medium with a lower trigger and a shorter wait. More false alarms.

**Settings** (Scan tab → CSI Motion Detection; most are under Advanced):

| Setting | What it means | Default |
|---|---|---|
| Movement needed before alerting | How long someone must move before you get an alert | 8 s |
| Stillness before all-clear | How long it must be quiet before the alert clears | 5 s |
| Devices that must agree | How many devices must see the movement at the same time | 2 |
| Trigger level | Lower catches smaller movement | set by sensitivity |
| Listen only, never transmit | The node never sends probes. Turn off only if it sees too little Wi-Fi | On |
| Include randomized-MAC devices | Also listen to phones and watches, not just routers | On |
| Min. seconds between mesh motion alerts | Wait at least this many seconds between motion alerts sent over mesh | 0 (no limit) |

**Getting false alarms with nobody there?** Turn on "Per-packet score to serial", watch the `sig` numbers while the room is empty, and set Trigger level just above the highest one.

> **CAN TRANSMIT**
>
> Off by default. If you turn off "Listen only, never transmit", the node sends Wi-Fi probe requests when there is too little traffic to measure. Anyone nearby can see them. Check local rules before turning it on.

Mesh commands: §11.

### 9.11 Packet Capture

Records raw traffic to SD as a standard pcap. Wi-Fi frames carry a radiotap header with channel, rate and RSSI. BLE captures are written as Bluetooth HCI, link type 187: the controller's own events, unmodified, so Wireshark shows LE Advertising Reports with address, address type, advertising data and RSSI. HCI carries no RF channel, so the capture does not report one.

Scan tab, pick **Packet Capture**, set radio and duration, press Start Capture. Files land in `/pcap` and list under the button, each with a download and a delete.

```
@ALL PCAP_START:0:300:0
@ALL PCAP_STOP
```

Arguments are radio, seconds, band. Radio 0 Wi-Fi, 1 BLE, one at a time; there is no both, and any other value, including 2, is taken as Wi-Fi. Unlike the `mode` field of the other scan commands, 2 does not mean both here. Band 0 is 2.4 GHz, 1 is 5 GHz, 2 both; 5 GHz needs a C5. Advanced sets a channel list, dwell, a management-frames-only filter, and the maximum file size.

The mesh reply names the radio that actually started, `PCAP_START: WIFI` or `PCAP_START: BLE`. Getting `WIFI` back from a command you meant as BLE means the radio field was wrong. A capture started by a vibration trigger or a Sentinel attack response is always Wi-Fi on 2.4 GHz; BLE is a manual choice only.

A capture stops on its own when the file reaches the size cap, when free space drops below the floor, or when the card stops accepting writes. The mesh reply carries the reason: `R=SIZECAP` or `R=WRITEFAIL`. The cap is 8 to 300 MB, 100 MB by default, set on the Scan tab or over mesh.

```
@ALL PCAP_LIMITS:150
@ALL PCAP_LIMITS
```

> **CAPTURES ARE LARGE AND CARRY EVERYTHING**
>
> A busy channel fills megabytes per minute. Check free space before a long run, and treat the file as a raw intercept.

> **STOP THE CAPTURE BEFORE YOU CUT POWER**
>
> FAT has no power-fail protection. Resetting or unplugging a node while a capture is writing can leave the card unreadable until it is reformatted, and the node then runs with no storage at all: no logging, no baseline, no captures. Stop the capture first.
>
> A node can rebuild its own card instead of waiting for someone with a card reader. This is off by default because rebuilding erases everything on the card.
>
> ```
> @ALL SD_REPAIR:ON
> @ALL SD_REPAIR:NOW
> @ALL SD_REPAIR:OFF
> ```
>
> `NOW` repairs once without changing the setting. With repair off, a failed mount says so once a minute and the SD line in Diagnostics shows how many times it has failed.

### 9.12 Scheduling scans

The Schedule panel on the Scan tab runs any scan at set times. The node runs it, so the web page can be closed.

1. Pick the scan in the top row and set its options.
2. Press **Schedule this scan**.
3. Set **Start**, **Duration** in minutes and **Repeat**, then press **Add**.

The block keeps the options that were set when you added it. Changing the scan's options later does not change the block. Delete it and add it again instead.

| Setting | Values |
|---|---|
| Start | Date and time in the node's local time, shown with its zone, for example MST |
| Duration | 1 to 1440 minutes |
| Repeat | Once, hourly, every 6 hours, every 12 hours, daily, weekly |

- **Time zone.** The node picks its zone from the GPS fix, daylight saving included. Daily and weekly blocks keep the same clock time across a daylight saving change. With no GPS fix, or outside the zone table, times are UTC.
- **Busy radio.** A block that comes due while another scan holds the radio runs as soon as the radio frees. A repeating block then moves to its next slot. If a mesh command or another start takes the radio at the same moment, the block is not lost; it tries again 5 seconds later.
- **Vibration and attack scans.** By default they wait for a scheduled scan: a vibration auto-scan is skipped, an attack response runs when the scan ends. Turn on **Interrupt scheduled scans** in Vibration Auto-Scan (System tab) or in the attack-response panel (Sentinel tab) to let that trigger stop the scheduled scan, run, and then resume the scheduled scan for its remaining time. A resumed Baseline starts its learning phase again. A scheduled block never takes the radio while a vibration or attack scan is waiting for it.
- **Over mesh.** `SCHED_ADD:2026-09-30T02:00|86400|/sniffer|detection=device-scan&deviceScanMode=2&secs=600` adds a daily block. `SCHED_LIST` prints the blocks to serial, `SCHED_DEL:2` deletes the second one.
- **Battery Saver.** Schedules pause while Battery Saver is on. A block that came due runs once when it ends.
- **Limits.** 8 blocks. They are kept in NVS and survive reboots.
- **Baseline blocks.** Adding one saves the baseline settings on the Scan tab to the node, and each run uses the baseline settings saved at that time.

> **CLOCK**
>
> Schedules need the clock from §4.3. On a node without a working RTC, blocks wait until the web UI is opened once after boot, which gives the node the time.

### 9.13 Reading the results

- **SD card, both builds.** Scan results in `/last_results.txt`, the probe database, vibration events in `/vibrations.jsonl`, packet captures in `/pcap`, sentinel incidents and the scan logs. Logging does not depend on which firmware you flashed.
- **Web UI, Full only.** The Results tab renders each scan as cards, and the API serves the same content as plain text.
- **Device database, Full only.** `/devicedb.jsonl` is what the Data Explorer browses. Headless does not write it.
- **Mesh.** Every alert above, plus `SCAN_DONE:` summaries carrying `TX=` broadcast and `DUP=` deduplicated counts.

> **EXPORTS CARRY LOCATION**
>
> SD logs, exports and mesh alerts can all contain GPS coordinates. Review any file before you post it publicly or attach it to a report. Privacy Mode only redacts the live web UI.

## 10 Vibration sensor and tamper response

### 10.1 How it works

The SW-420 is a mechanical vibration switch with a comparator and a trimpot on board. It has one job: pull its digital output high when movement exceeds the threshold the trimpot sets. That output goes to GPIO 2 on the ESP32,

configured as a pull-down input with a rising-edge interrupt.

When it fires, the firmware:

- Sends a mesh alert if vibration alerts are enabled: `NODE: VIBRATION: Movement detected at TIME [GPS:lat,lon]`
- Writes a structured event to `/vibrations.jsonl` on the SD card
- Optionally starts a scan (§10.3)
- Counts toward the tamper auto-erase trigger if that is armed (§10.4)

Alerts are rate-limited to one every 3 seconds, so continuous shaking produces a steady trickle rather than a flood. The System diagnostics show `Vibration sensor: Standby` or `Vibration sensor: Active – Last: TIME (Ns ago)`.

Turn alerts on and off with `@AH01 VIBRATION_ON` and `VIBRATION_OFF`, or the toggle in the web UI's Sensor Alerts card. `VIBRATION_STATUS` reports the current state.

### 10.2 Tuning the trimpot

The trimpot is the small brass screw on the sensor module. It sets the sensitivity threshold, and the correct setting depends entirely on where the node is mounted. A pole in the wind is not a shelf indoors. Tune it **after** the node is in its final position, not on the bench.

1. Power the node and watch the sensor's own indicator LED, or open a serial monitor and watch for `[VIBRATION]` lines.
2. Leave the node completely still. If it is triggering by itself, turn the screw an eighth of a turn at a time in the direction that stops it. Wait several seconds between adjustments. The module reacts slowly.
3. When it is silent at rest, back off slightly further so ambient movement does not set it off.
4. Now tap the enclosure the way a person handling it would. It should fire.
5. If it does not, turn back the other way a fraction at a time until a deliberate tap triggers it but the node at rest does not.
6. Confirm the events actually reached the firmware: the diagnostics timestamp updates, and each event is appended to `/vibrations.jsonl`.

> **TUNING BY SYMPTOM**
>
> - **Constant false alerts.** Too sensitive, or mounted on something resonant. Reduce sensitivity, and consider a foam pad or a stiffer mount.
> - **Nothing fires even when you shake it.** Too insensitive, or the sensor is not seeing GPIO 2. Check the diagnostics line and the solder joints on the sensor header.
> - **Fires only sometimes.** You are on the edge of the threshold. Back it off a fraction toward sensitive.

### 10.3 Vibration auto-scan

Movement can start a scan automatically. This suits a node used as a trip sensor rather than a continuous monitor.

```
@AH01 VIBSCAN_SET:1:2:60:60:1  # on : mode : secs : cooldown s : interrupt schedule
@AH01 VIBSCAN_STATUS
```

| Mode | Scan started | Mode | Scan started |
|---|---|---|---|
| 0 | None | 4 | List scan (targets) |
| 1 | All-device scan | 5 | Drone / Remote ID |
| 2 | Probe-request detect | 6 | Deauth (blue team) |

| Mode | Scan started | Mode | Scan started |
|---|---|---|---|
| 3 | Randomized-MAC detect | 7 | Baseline anomaly |
| | | 8 | Packet capture to SD |

Duration 0 means run until stopped. The trigger is skipped if a scan is already running or the node is in battery saver. Mode 8 writes an `auto_` capture, which is subject to the auto-capture budget rather than kept forever.

### 10.4 Tamper auto-erase

The node can destroy its own data when it is moved. This is **off by default**, and it is permanent and irreversible.

| Parameter | Range | What it does |
|---|---|---|
| Setup delay | 30 s to 10 min | Grace period after arming, so installing the node does not wipe it |
| Vibrations required | 2 to 5, default 3 | How many movement events count as tampering |
| Detection window | 10 to 60 s | The period those events must fall within |
| Erase delay | 10 to 300 s | Countdown before destruction, your window to cancel |
| Cooldown | 5 to 60 min | Minimum time between tamper attempts, to blunt false triggers |

```
@AH01 AUTOERASE_ENABLE:60:30:3:30:300:<answer>   # setup:erase:vibs:window:cooldown:answer
@AH01 AUTOERASE_STATUS
@AH01 AUTOERASE_DISABLE:<answer>
@AH01 ERASE_CANCEL:<answer>                       # abort a running countdown
```

During the setup period the node still reports movement, so you can watch the arming from a distance: `VIBRATION: Movement in setup mode (active in Ns)`. When it triggers you get `TAMPER_DETECTED: Auto-erase in Xs`.

#### Remote erase

Every mesh command marked `<answer>` takes a one-time answer to a token. The erase key (§8.1) never goes on the mesh for these commands, so a captured command can't be replayed:

1. Send `@AH01 ERASE_REQUEST` (one per 20 s). The node replies `AH01: ERASE_TOKEN:AH_3F2A91C0_77B0E412_000004D2 Expires:300s`.
2. Sign the token: `printf '%s' 'AH_3F2A…' | openssl dgst -sha256 -hmac 'yourKey'`.
3. Within 300 s send the 64 hex characters, or the first 8, as `<answer>`: `@AH01 ERASE_FORCE:<answer>`. A token takes one attempt, right or wrong.

The key itself is refused over mesh. 5 wrong answers, or 5 wrong keys on the web, lock erase commands for 10 min. Logs show keys and answers as ∗∗∗.

In the web UI, enter the key in the Authorization field.

After a wipe the node plants a dummy IoT weather configuration in place of its own, so a recovered device does not obviously read as a sensor.

#### Factory reset

```
@AH01 FACTORY_RESET:FULL:<answer>    # FULL | CONFIG | DATA
```

Single node only.

> **DATA DESTRUCTION IS IRREVERSIBLE**
>
> Test your settings with a node that holds nothing you care about. A wipe cannot be undone, and the countdown is short by design.

# 11 Mesh command reference

Send these as ordinary text messages from the Meshtastic app, a Command Center, or any client on your channel. `@ALL` broadcasts. A node ID targets one node. Times are the node's local time once GPS fixes the zone, UTC before that.

### Core

| Command | Description |
|---|---|
| `STATUS` | Mode, scan state, hits, temperature, uptime, GPS |
| `STOP` | Stop all operations and any pending attack response, flush pending mesh transmissions |

### Configuration

| Command | Does | Parameters |
|---|---|---|
| `CONFIG_TARGETS` | Set the watchlist | Pipe-delimited MACs, OUIs or SSIDs |
| `CONFIG_NODEID` | Rename the node | 2 to 5 alphanumeric characters |
| `CONFIG_RSSI` | Set the RSSI floor | −128 to −10 |
| `CONFIG_CHANNELS` | Set the channels to sweep | List or range, e.g. `1..11` |
| `CONFIG_BAND` | Pick the band, C5 only | 0 = 2.4 GHz, 1 = 5 GHz, 2 = both |
| `CONFIG_DEDUP_TTL` | Set cross-scan MAC dedup | 0 to 3600 s, 0 disables |
| `CONFIG_SESSION_DEDUP` | Toggle per-session dedup | 0 or 1 |
| `MESH_DEDUP_CLEAR` | Clear the dedup cache | None |
| `DEVICE_DB_CLEAR` | Clear the discovered-device DB, headless, Beta and C5 | None |
| `CONFIG_ERASE_PSK` | Set the erase key | `<new>:<current>`, 8 to 64 characters, no `:`; §8.1 |

### Scanning

| Command | Does | Parameters |
|---|---|---|
| `SCAN_START` | Hunt the watchlist | `mode:secs:channels[:FOREVER]` |
| `DEVICE_SCAN_START` | List everything in range | `mode:secs[:FOREVER[:+PROBE]]` |
| `PROBE_START` and `PROBE_STOP` | Collect probe requests | `mode:secs[:FOREVER][:+ALL]` |
| `BASELINE_START` and `BASELINE_STATUS` | Learn the area, then flag changes | `duration[:FOREVER]`, min 60 s |

| Command | Does | Parameters |
|---|---|---|
| BASELINE_WATCH | Watch for changes from now | None; baseline must be running |
| DRONE_START | Watch for drone Remote ID | secs[:FOREVER] |
| DEAUTH_START | Watch for deauth attacks | secs[:FOREVER] |
| RANDOMIZATION_START | Link randomized MACs to devices | mode:secs[:FOREVER] |
| CSI_MOTION_START | Detect movement in the room, Beta and C5 | secs[:FOREVER][:CH<n>][:TELEM][:RAW][:GAP<s>][:LISTEN_ONLY\|ALLOW_TRANSMIT][:MGMTONLY\|MGMTDATA][:SOLICIT<ms>] |
| CSI_CFG | Tune the motion trigger, Beta and C5 | trigger:hold_ms:consec:channel or SENSITIVITY=LOW\|MEDIUM\|HIGH, MIN_MOTION=, CLEAR_AFTER=, SPOTS=, BROADCAST=ON\|OFF, ALLOW_RANDOM=ON\|OFF |
| CSI_STATUS and CSI_JSON | Dump motion state to serial, Beta and C5 | None |
| CSI_RECAL | Drop a saved trigger, Beta and C5 | None |
| CSI_EXCLUDE | Ignore one MAC for motion, Beta and C5 | <MAC> or NONE |
| MESH_TX_CANCEL | Drop queued mesh traffic, keep scanning | None |
| PCAP_START and PCAP_STOP | Record traffic to SD as pcap | radio:secs:band[:CH<list>] |
| PCAP_LIMITS | Set or read the capture size cap | [MB], 8 to 300 |
| PCAP_AUTO | Set or read the auto-capture SD budget | [budgetMB,floorMB] |
| SD_REPAIR | Rebuild an unmountable SD card; erases it | ON\|OFF\|NOW |
| TRIANGULATE_START | Locate a MAC across nodes | target:duration[:rfEnv[:wifiPwr:blePwr]] |
| TRIANGULATE_RESULTS and TRIANGULATE_STOP | Report the fix, or stop | None |

Mode is 0 Wi-Fi, 1 BLE, 2 both. CH0 or no channel surveys and pins the busiest. PCAP is the exception: its radio field is 0 Wi-Fi / 1 BLE only, one radio per capture, and 2 is taken as Wi-Fi rather than both. PCAP band 0 is 2.4 GHz, 1 is 5 GHz, 2 both; 5 GHz needs a C5.

## Sentinel

| Command | Does | Parameters |
|---|---|---|
| SENTINEL_ON, SENTINEL_OFF, SENTINEL_STATUS | Start, stop, report | None |
| SENTINEL_MODE | Pin a channel or hop | defend or scan |
| SENTINEL_BOOT | Auto-start at power-on | 1 or 0 |
| GROUP | Toggle a detector group | <dos\|rogue\|recon\|physical\|mesh\|all>: <on\|off> |
| DETECT_CFG and DETECT_CFG_GET | Set or dump detector tunables | JSON, 180 characters or fewer |

| Command | Does | Parameters |
|---|---|---|
| INCIDENTS and INCIDENTS_CLEAR | Dump or clear the incident log | [:1–200] |
| ATTACKER_TRILAT and ATTACKER_TRILAT_STATUS | Triangulate a confirmed attacker, or report the setting | 1 or 0, off by default |
| DETECT_COUNTS | Entry count per detector, pending attack responses | None |

### Security, power and heartbeat

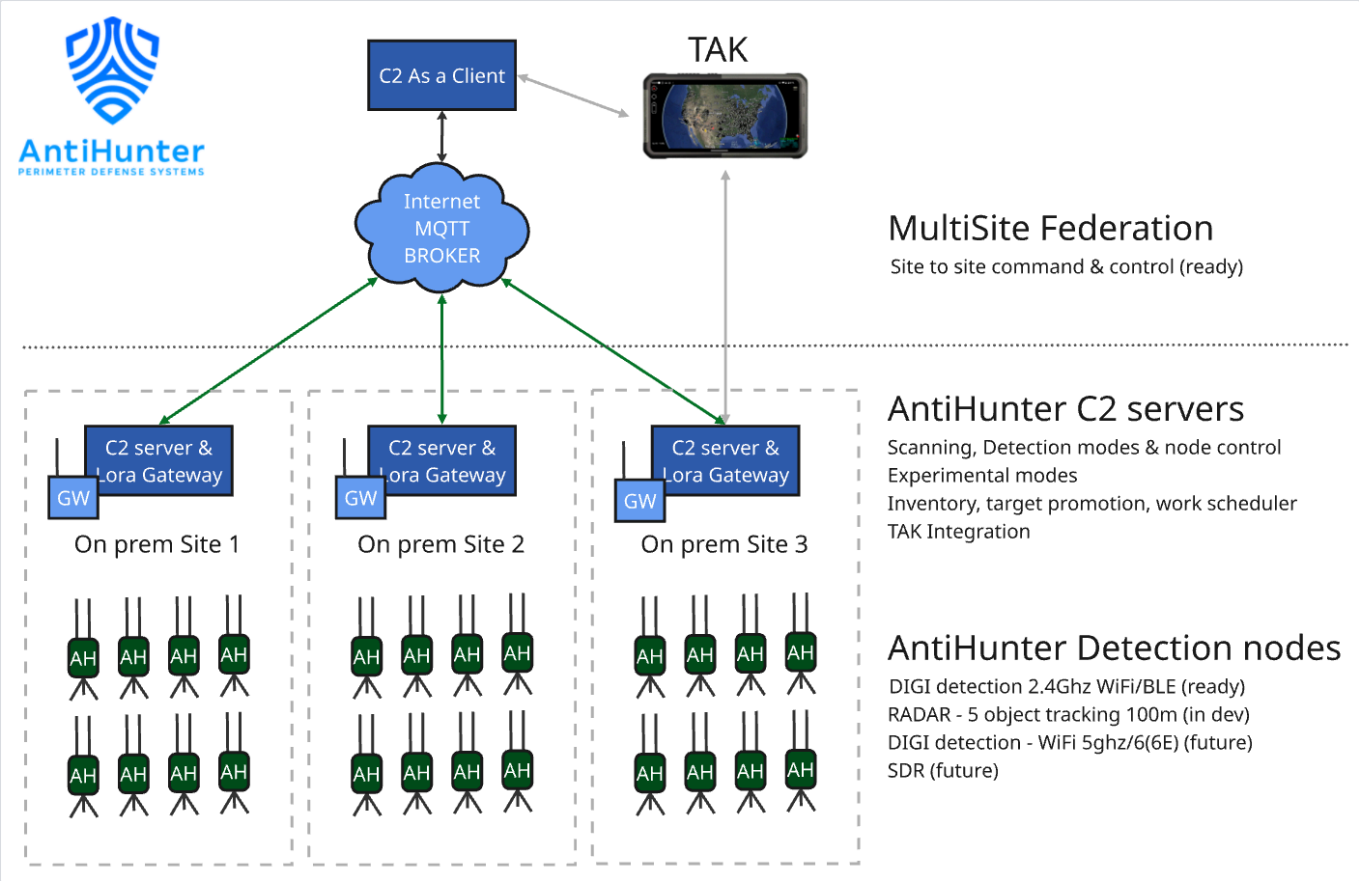
| Command | Does | Parameters |
|---|---|---|
| ERASE_REQUEST, ERASE_FORCE, ERASE_CANCEL | Token, wipe, abort | None, <answer>, <answer>; §10.4 |
| AUTOERASE_ENABLE | Wipe if the node is moved | setup:erase:vibs:window:cooldown:answer |
| AUTOERASE_DISABLE and AUTOERASE_STATUS | Turn that off, or report it | <answer>, none |
| VIBRATION_ON, VIBRATION_OFF, VIBRATION_STATUS | Enable, disable, report the sensor | None |
| VIBSCAN_SET and VIBSCAN_STATUS | Start a scan when moved | en:mode:dur[:cooldown[:interrupt]] |
| SCHED_ADD, SCHED_LIST, SCHED_DEL | Add, list or delete a schedule block | at\|repeat s\|path\|body, none, block number |
| FACTORY_RESET | Reset one node | <FULL\|CONFIG\|DATA>:<answer> |
| BATTERY_SAVER_START | Drop to low power | Minutes, 1 to 30, default 5 |
| BATTERY_SAVER_STOP and BATTERY_SAVER_STATUS | Return to normal, or report | None |
| HB_ON, HB_OFF, HB_INTERVAL | Periodic status broadcast, off by default | Interval 1 to 60 min |

### Ten commands to learn first

```
@ALL STATUS                          # who is alive, and what are they doing
@ALL STOP                            # stop everything, everywhere
@AH01 CONFIG_NODEID:GATE1            # name a node
@ALL CONFIG_TARGETS:AA:BB:CC|MyWiFi  # set the watchlist
@ALL SCAN_START:2:600:1..11          # 10 min watchlist scan, Wi-Fi + BLE
@ALL DEVICE_SCAN_START:2:300         # survey everything nearby
@ALL BASELINE_START:900              # learn the site for 15 min
@ALL SENTINEL_ON                     # attacker-tool detection
@ALL DRONE_START:0                   # drone watch until stopped
@AH01 BATTERY_SAVER_START:10         # idle a node without losing the mesh
```

# 12 Command Center

One node needs no server. A fleet benefits from one: AntiHunter Command & Control PRO aggregates every node's detections onto a live map, with TAK, MQTT and ADS-B integration and a Sentinel console.




Nodes operate independently and coordinate over the mesh. The Command Center aggregates.

### 12.1 What you need

- A computer to run it on: Linux, macOS or Windows 10/11. It can be the same machine you flash from.
- A Meshtastic radio on USB, joined to **your** channel, to receive the mesh.
- Node.js 20 or newer and PostgreSQL. The installer puts both in place for you.

### 12.2 Quick install: Linux and macOS

```
curl -o setup-local.sh \
  https://raw.githubusercontent.com/TheRealSirHaXalot/AntiHunter-Command-Control-
PRO/main/scripts/setup-local.sh
chmod +x setup-local.sh
./setup-local.sh
```

Run it as a normal user, **not** with sudo. It asks for sudo itself when it needs it. It installs the prerequisites, clones the project, sets up the database, runs the migrations and seeds an admin login. When it finishes, open **http://localhost:5173** and log in with the credentials it printed.

### 12.3 Quick install: Windows 10/11

1. Install Git in any PowerShell window, then close it so Git lands on your PATH:

```
winget install -e --id Git.Git --accept-package-agreements --accept-source-agreements
```

2. Open PowerShell **as Administrator**. The installer needs it to install system software.
3. Clone and run the installer:

```
cd $HOME
git clone https://github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO.git
cd AntiHunter-Command-Control-PRO
powershell -ExecutionPolicy Bypass -File scripts\setup-windows.ps1
```

4. Press Enter to accept the defaults. The first run takes a few minutes. **Write down the admin email and password it shows you.**
5. Start it with **Start-AntiHunter.cmd** in the project folder, or `pnpm AHCC`.
6. Open **http://localhost:5173** and log in.

> **WINDOWS NOTES**
>
> - The firewall prompt on first start: click **Cancel** if you only use it on this PC, since localhost works either way. To reach it from other devices, click Allow and tick **Private** only.
> - For serial hardware, put `SERIAL_DEVICE=COM3`, your port from Device Manager → Ports, in `apps\backend\.env` and restart.
> - Do not use WSL. COM ports do not pass through cleanly.

### 12.4 Docker

Needs Docker 25 or newer. If you have no serial adapter at `/dev/ttyUSB0`, which is every macOS and Windows host, comment out the `devices:` and `group_add:` lines under the `backend` service first, or the container will not start.

```
git clone https://github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO.git
cd AntiHunter-Command-Control-PRO
docker compose up -d --build
```

Three containers come up: Postgres, the backend on port 3000, and the frontend on **http://localhost:8080**. The seeded admin defaults to `admin@example.com` and `admin`. Logs: `docker compose logs -f backend`. Stop: `docker compose down`, adding `--volumes` to delete the database too.

> **CHANGE THE ADMIN PASSWORD IMMEDIATELY**
>
> The seeded credentials are published in the documentation. Change them at first login, before the server is reachable by anything but you. Enable two-factor authentication and HTTPS if the instance will be reached across a network, and work through the hardening checklist in the project README.

### 12.5 Updating

Use the in-app update page, or `git pull` in the project folder. Docker: `git pull && docker compose up -d --build`. The backend re-applies migrations on boot.

### 12.6 Reaching it from anywhere, and alerts to your phone

The Command Center's Remote Connections guide walks through each option with a check at every step:

| I want to… | Use |
|---|---|
| Run the map and console from my phone, away from site | Tailscale (private VPN between your own devices) |
| Let others in without installing an app | Cloudflare Tunnel with an email login |
| Get a message when a node detects something | Push, Signal, ntfy, Matrix, Discord, IFTTT, Home Assistant, email |
| See nodes and targets in ATAK, iTAK or WinTAK | TAK bridge |
| Feed detections to scripts, or link two sites | MQTTS broker and federation |

Never forward the Command Center's ports on your router. Set its admin password and two-factor first; the guide starts there.

# 13 Deployment

## 13.1 Siting

- **Height helps more than power.** Get the node above metal, vehicles and crowds. A few metres of elevation beats any antenna change.
- **Antennas vertical**, and clear of the enclosure's own metalwork and of each other.
- **GPS wants sky.** The helix or patch needs a clear upward view. Indoors it may never lock, and without a fix your detections carry no coordinates and triangulation cannot run.
- **Keep LoRa and Wi-Fi antennas apart** as far as the enclosure allows.
- For multi-node triangulation, spread nodes around the area rather than in a line, and give each one a clear GPS fix. Matched antennas on every node.



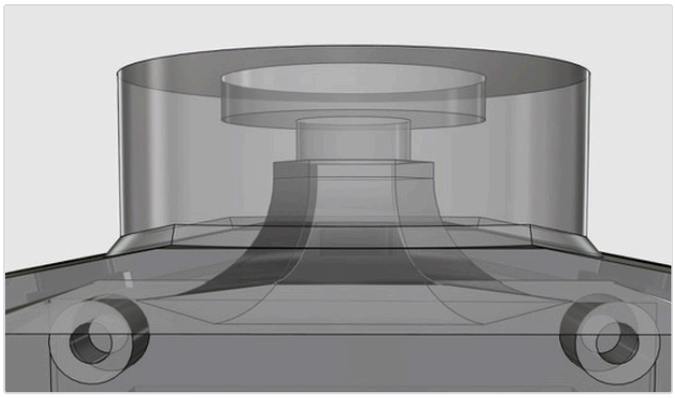
Mast, tripod, can and puck builds. The enclosure changes. The node does not.

## 13.2 Sealing and mounting

- Fit the TPU seals for the housing, GPS and USB-C port. Silicone grease on the seals and SMA threads helps in persistent wet.
- No IP rating is claimed for the printed enclosure. Treat it as weather-resistant, not waterproof, and site it accordingly.
- The base takes a ¼"-20 insert for tripods and the universal bracket. STL files for brackets, solar mounts and radar extensions are in the `hw/` folder.
- Point the vented lid and fan opening downward or shielded so water does not sit in them.




An enclosed build with the antennas vertical and the switch centered.



Vented lid, SMA bulkheads on top, battery housing below.

### 13.3 Power planning

- Roughly 160 mA idle, 600 mA scanning. Aggressive presets and continuous promiscuous capture sit at the top of that range.
- Battery Saver stops scanning, drops the CPU to 80 MHz and polls GPS once a minute while keeping the mesh alive. That is the right state for a node that only needs to be reachable.
- Solar mount frames are in the STL folder. Feed solar through the UPS board, never directly to the PCB.
- Never use the chassis USB input and the UPS fast-charge input at the same time.

### 13.4 Playbooks

Set-and-forget setups for common situations. Schedule blocks run with the web page closed and survive reboots; set them in the Scan tab (pick the scan, then **Schedule this scan**) or over mesh with `SCHED_ADD` (full firmware). Times are the node's local time. Repeat `86400` is daily, `604800` weekly, `0` once.

#### Overnight perimeter watch (home, shop, site)

Learn what belongs, then flag anything new after dark. Pair with Sentinel for attacks on your Wi-Fi.

```
@AH01 SCHED_ADD:2026-10-05T21:00|86400|/sniffer|detection=baseline&secs=32400   # 9 h, nightly
@AH01 SENTINEL_BOOT:1                    # attack detection from power-on
@AH01 VIBRATION_ON
```

You get `ANOMALY-NEW` for a device that was not there at dusk, and Sentinel alerts for deauth, evil twins and similar attacks.

#### Someone casing the place

The same phone or car showing up on different days is the tell. Run a device scan while they are around, put the MACs you care about on the watchlist, then hunt them on a schedule.

```
@AH01 DEVICE_SCAN_START:2:600
@ALL CONFIG_TARGETS:AA:BB:CC:DD:EE:FF|11:22:33
@ALL SCAN_START:2:0:1..11:FOREVER                                  # or schedule it:
@AH01 SCHED_ADD:2026-10-05T08:00|86400|/scan|mode=2&secs=36000&ch=1..11
```

A hit sends `Target: MAC RSSI Type Name GPS V=vendor` at once; the web UI shows Getting closer or Moving away as it moves.

#### While you are away

Start a baseline when you arrive. When you leave, press **Watch for changes from now** or send `@AH01 BASELINE_WATCH`. Back home, the results list what is new, what is gone, and what moved closer.

#### Drones over the property

```
@AH01 SCHED_ADD:2026-10-05T06:00|86400|/drone|droneScanMode=2&secs=57600   # daylight hours
```

Each aircraft broadcasting Remote ID is reported once with its ID, position and operator location when sent.

#### Your Wi-Fi is being attacked

Most Wi-Fi attacks start by knocking clients off with deauthentication frames: to capture a handshake, push them onto an evil twin, or just jam. The Deauth scan catches that on every build; run it around the clock.

```
@AH01 DEAUTH_START:0:FOREVER
@AH01 SCHED_ADD:2026-10-05T00:00|86400|/sniffer|detection=deauth&secs=86400   # or as a daily
block
@AH01 SENTINEL_ON                          # also evil twins, karma, PMKID, floods
```

Alerts name the source MAC and target. Sentinel can also start triangulation, a capture or a sweep on a confirmed attack (Sentinel tab → On confirmed attack).

#### Unattended in a vehicle or camp

```
@AH01 VIBSCAN_SET:1:1:300:600              # a bump starts a 5 min device scan
@AH01 BATTERY_SAVER_START:10                # mesh stays reachable between events
```

#### Several sites, one screen

Run the Command Center (§12) on a machine at one site with a Meshtastic radio on USB. Reach it from your phone over Tailscale and route alerts to Signal or push (§12.6).

### 13.5 Before you walk away

- ☐ AP credentials changed, or Headless firmware flashed
- ☐ LoRa region set and your own encrypted channel primary, public channel off
- ☐ Node ID set and recorded
- ☐ Clock set, GPS has a fix
- ☐ SD card mounted and writable
- ☐ Vibration sensor tuned in its final position
- ☐ Auto-erase parameters understood if you armed it, and the erase key recorded somewhere safe
- ☐ The node answers `@NODEID STATUS` over the mesh from where you will be standing
- ☐ Antennas tight, seals seated, lid closed

# 14 Maintenance and troubleshooting

## 14.1 Symptom table

| Symptom | Likely cause | What to do |
|---|---|---|
| No `Antihunter` Wi-Fi network | Headless firmware flashed, or the AP name and password were changed | Check over serial which build is running. Reflash Full if you need the UI. |
| Web UI will not load | Not connected to the node's own AP, or using https | Join the node's AP, then open **http://192.168.4.1** exactly. |
| Flashing fails or the port never appears | Charge-only USB cable, a serial monitor still holding the port, both USB ports connected, or the wrong browser | Use a data cable, close other serial tools, unplug the radio, use Chrome or Edge. |
| SD card not detected | Not FAT32, a 32 GB or larger card, or not seated | Reformat FAT32 on a card under 32 GB. Reseating means opening the enclosure. |
| No GPS fix | Antenna not connected, or no view of the sky | Check the U.FL or SMA connection, move outdoors, give it time. The module runs hot while searching, which is normal. |
| Nothing arrives over the mesh | Region `UNSET` so the radio is receive-only, serial module off, wrong pins for the board, or nodes on different channels or keys | Re-run `meshtastic_config.py` and print the settings. Confirm every node carries the same channel. |
| Node hears commands but never answers | Region unset on *that* node, so it cannot transmit | Set its region. |
| Constant vibration alerts | Trimpot too sensitive, or a resonant mount | Retune in place (§10.2), pad the mount. |
| Vibration never fires | Trimpot too insensitive, or a bad joint on the sensor header | Retune, check the diagnostics line and the solder joints. |
| Node scans forever | A scan was started with `:FOREVER` or duration 0 | `@ALL STOP`, or the Stop button in the UI. |
| Settings gone after flashing | Erase Device was ticked | Reconfigure. Settings mirror to the SD card and self-heal, but a deliberate erase clears NVS. |
| Node hot to the touch | Normal under sustained scanning | Fit the fan and thermal switch, ventilate the enclosure, consider a relaxed preset. |
| Detections with no coordinates | No valid GPS fix at the time | Expected behavior. Position is only attached when the fix is valid. |

## 14.2 Routine maintenance

- Pull the SD card periodically and archive the logs, then clear them, so a lost node carries less.
- Check the cells. A swollen or damaged 18650 is retired immediately, never recharged.
- Re-check antenna tightness and seal condition after any period in the field.
- Reflash on the same channel to pick up fixes. Read the release notes first.

### 14.3 Getting help

Issues and Discussions at **github.com/lukeswitz/AntiHunter**. A useful report includes: firmware build and channel, Full or Headless, board type, what you did, what you expected, what happened, and the serial output around the failure. Scrub coordinates and MAC addresses from anything you paste. See §15.

## 15 Legal and safety

### 15.1 Authorization

AntiHunter is distributed for lawful, authorized defensive use only: security research, blue-team training, compliant monitoring, event security and network auditing. Operate it only on infrastructure, networks, devices, spectrum and datasets you own or have explicit written permission to assess. The detection, correlation and triangulation features exist to characterize an environment you are authorized to assess, not to identify, follow or profile people. Targeted surveillance, stalking, harassment and tracking individuals without informed consent are prohibited.

### 15.2 Radio compliance

You are responsible for every rule governing radio use where you operate: FCC Part 15 and Part 97, CE/RED, Ofcom and their national equivalents; ISM band allocations, power limits and duty cycles; and any licensing conditions for your class of operation. LoRa is region-locked, 868 MHz EU, 915 MHz US, 923 MHz Asia, and running a region outside the one you are in may be unlawful. Never use the node to cause harmful interference.

### 15.3 Interception and privacy

Passive reception of radio emissions is regulated separately from network access in most jurisdictions. Capturing, storing, decoding or disclosing frame contents and identifiers may engage the US Wiretap Act and ECPA, state two-party-consent statutes, the UK Investigatory Powers Act or equivalents. Determine the lawfulness of each capture mode where you are before enabling it.

MAC addresses, probe request contents, BLE advertisements and Remote ID broadcasts may be personal data under GDPR, UK GDPR, CCPA/CPRA and similar regimes. Collect only with a lawful basis, minimize what you keep, set a retention and destruction schedule, and honor data-subject rights. The maintainers never receive, process or host your data.

### 15.4 Drone Remote ID

Receiving broadcast Remote ID is permitted in most places. Using that data to locate, approach, confront or interfere with an aircraft or its operator may violate aviation, harassment or anti-stalking law, including 18 U.S.C. § 32. Reception is not authorization to act.

### 15.5 Hardware and battery safety

Kits, bare boards and assembled units are supplied for people competent in electronics assembly and operation. Correct assembly, ESD control, antenna selection and attachment, and supply of regulated 5 V power are your responsibility. Operating the transceiver without a properly matched antenna can damage the hardware. Lithium cells are never supplied. Sourcing, protection, charging, storage, transport and disposal are yours, and carry fire and injury risk. The hardware is not certified for and must not be used in life-safety, medical, aviation, automotive or industrial-control applications. It is not a substitute for a certified security or alarm system, and its output is advisory only, subject to false positives and false negatives.

### 15.6 Experimental features

Modes marked beta or experimental, including Sentinel, Triangulation and MAC Randomization Correlation, are unvalidated, may produce inaccurate or misleading output, and must not be relied on for operational, evidentiary, investigative or safety decisions.

> **DISCLAIMER**
>
> THE PROJECT, INCLUDING SOFTWARE, FIRMWARE, HARDWARE DESIGNS AND DOCUMENTATION, IS PROVIDED "AS IS" AND "AS AVAILABLE", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, TITLE, NON-INFRINGEMENT, ACCURACY, DETECTION EFFICACY OR UNINTERRUPTED OPERATION. This supplements and does not limit the warranty disclaimer and liability limitation in sections 15 to 17 of the GNU Affero General Public License v3.0.
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# Appendix A · Pinout

XIAO ESP32-S3 on the DIGI NODE board. Pin assignments may change with board revisions. Verify against your revision.

| Function | GPIO | Notes |
|---|---|---|
| Vibration sensor | 2 | SW-420 tamper detection, rising-edge interrupt, input pull-down |
| RTC SDA | 3 | DS3231 I²C data |
| RTC SCL | 6 | DS3231 I²C clock |
| GPS RX | 44 | NMEA in |
| GPS TX | 43 | Unused |
| SD CS | 1 | Chip select |
| SD SCK | 7 | SPI clock |
| SD MISO | 8 | SPI MISO |
| SD MOSI | 9 | SPI MOSI |
| Mesh RX | 4 | From the Meshtastic radio |
| Mesh TX | 5 | To the Meshtastic radio |

Radio side: serial module in TEXTMSG mode at 115200 baud, pins 19 RX / 20 TX on Heltec V3, 10 RX / 9 TX on T114.

# Appendix B · Bill of materials

### Core

- 1× AntiHunter DIGI PCB, 82 mm, 2-layer
- 1× Seeed XIAO ESP32-S3, 8 MB flash minimum, or XIAO ESP32-C5 for 2.4 and 5 GHz
- 1× Heltec Wi-Fi LoRa 32 V3.2, T114 also compatible
- 1× ATGM336H GPS module
- 1× DS3231 RTC module
- 1× SW-420 vibration sensor
- 1× micro SD reader module
- 1× SD card, 8 GB supplied on every tier but the Bare PCB. FAT32, 32 GB and larger not recommended
- 1× KSD9700 normally-open thermal switch, 40 °C

### Connectors and fasteners

- 6× JST 2.54 2-pin terminals
- 3× JST power male cables for the switch, thermal switch and power board
- 10× M3 heat-set inserts
- 4× M2 heat-set inserts for the power board
- 2× M3 15 mm brass standoffs
- M3×4-6mm screws for the PCB and enclosure lids, M2×4-6mm for the power board
- 2× M2.5 screws, supplied with the fan
- 1× ¼"-20 tripod insert

### Antenna and cabling

- 3× U.FL to SMA pigtail, 10 to 20 cm
- SMA bulkheads
- 1× 6 dBi 2.4 GHz antenna
- 1× 6 dBi LoRa antenna for your region
- 1× GPS antenna. Passive ceramic patch on the kit and populated PCB, active helix on the assembled unit

### Power and thermal

- 1× 30 mm 5 V fan with JST lead
- 1× 3-pin mini on/off switch
- 1× waterproof Type-C female chassis jack, 2-pin 22 AWG leads, 14 mm panel nut, dust cap
- 1× Type-C 15 W 3 A 5 V UPS board, dual 18650
- 1× 2 A fast-blow fuse, recommended, not supplied
- 2× 18650 cells, protected flat-top, never supplied with any tier

### Enclosure

- 3D-printed weatherproof enclosure. STLs, seals, brackets and solar mounts in `hw/Prototype_STL_Files/`

Part links and photographs: `hw/Prototype_STL_Files/BOM-Links.md`

# Appendix C · Defaults, addresses and files

| Item | Default | |
|---|---|---|
| Wi-Fi AP name | `Antihunter` | **change it** |

| Item | Default | |
|---|---|---|
| Wi-Fi AP password | `antihunt3r123` | **change it** |
| Wi-Fi AP hidden | off, SSID beacons | optional, see §8.1 before turning it on |
| Web UI | `http://192.168.4.1` | Full firmware only |
| Erase key | not set | **set it** if using erase or factory reset |
| Meshtastic serial | TEXTMSG, 115200 | pre-configured on tiers 1 and 3, you set it on 2 and 4 |
| LoRa region | `UNSET`, receive only | **set it** |
| Meshtastic channel | public default | **replace with your own encrypted channel** |
| Scan preset | Balanced | |
| Wi-Fi channels | 1 to 11 | |
| Mesh dedup TTL | 300 s | 0 disables |
| Vibration alerts | enabled | rate-limited to one per 3 s |
| Vibrations required for tamper | 3 | range 2 to 5 |
| Auto-erase | disabled | |
| Heartbeat | disabled | |
| Karma bait | off | the only transmitting detector |
| Attacker auto-triangulation | off | |
| C2 frontend | `http://localhost:5173`, Docker 8080 | |
| C2 seeded admin | `admin@example.com` and `admin` | **change at first login** |

### Files on the SD card

| Path | Contents |
|---|---|
| `/last_results.txt` | Most recent scan results as text. Written by both builds |
| `/devicedb.jsonl` | Device database. Full firmware only, browsed by the Data Explorer |
| `/vibrations.jsonl` | Structured vibration events |
| `/pcap/wifi_*.pcap` and `/pcap/ble_*.pcap` | Packet captures you started by hand. Never pruned automatically |
| `/pcap/wifi_auto_*.pcap` and `/pcap/ble_auto_*.pcap` | Captures started by a vibration trigger or a Sentinel attack response. Oldest are deleted to stay inside the budget and keep the card's free-space floor |

# Appendix D · Glossary

| | |
|---|---|
| **BSSID** | The MAC address of an access point. |
| **OUI** | The first three bytes of a MAC address, identifying the manufacturer. Watchlisting an OUI matches a whole vendor. |
| **Probe request** | A frame a device sends looking for networks it knows. Reveals remembered SSIDs. |

| | |
|---|---|
| **Ghost SSID** | A probed network with no matching AP responding nearby. |
| **Evil twin** | A rogue AP impersonating a legitimate network to capture clients. |
| **Karma** | An AP that answers every probe request, claiming to be whatever a client asks for. |
| **Deauth** | A management frame that disconnects a client. Used to force reconnections and capture handshakes. |
| **PMKID** | A value in the first EAPOL message that can be captured and cracked offline without a full handshake. |
| **Remote ID** | The broadcast identification a drone emits under ASTM F3411 and equivalent rules. |
| **RSSI** | Received signal strength, in dBm. Less negative is stronger. |
| **NVS** | Non-volatile storage on the ESP32 where settings persist across reboots. |
| **SoftAP** | The Wi-Fi access point the Full firmware hosts for its web UI. |
| **U.FL / IPX** | The miniature snap-on RF connector on each module. |

# Appendix E · Quick reference

Print this page and keep it with the node.

### Never

- Power from anything but a regulated 5 V source
- Power the radio with no LoRa antenna fitted
- Connect both USB ports at once
- Press a U.FL connector at an angle
- Leave the default AP password or the public mesh channel in place

### Addresses

| | |
|---|---|
| Web UI | `http://192.168.4.1`, AP `Antihunter` |
| Web flasher | `lukeswitz.github.io/AntiHunter` |
| Firmware, docs, issues | `github.com/lukeswitz/AntiHunter` |
| Command Center | `github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO` |
| Meshtastic flasher | `flasher.meshtastic.org` |

### Ten commands

```
@ALL STATUS                          who is alive
@ALL STOP                            stop everything
@AH01 CONFIG_NODEID:GATE1            name a node
@ALL CONFIG_TARGETS:AA:BB:CC|MyWiFi  set the watchlist
@ALL SCAN_START:2:600:1..11          watchlist scan, 10 min
@ALL DEVICE_SCAN_START:2:300         survey everything
@ALL BASELINE_START:900              learn the site
@ALL SENTINEL_ON                     attacker-tool detection
@ALL DRONE_START:0                   drone watch until stopped
@AH01 BATTERY_SAVER_START:10         idle, mesh stays up
```

### Setup order for a new node

1. Fit the antennas
2. Flash AntiHunter, Full to start with
3. Change the AP credentials
4. Set the radio region
5. Create your encrypted channel and disable the public one
6. Set the node ID and the clock
7. Confirm the SD card and GPS
8. Tune the vibration trimpot in place
9. Verify `@NODEID STATUS` answers over the mesh
10. Seal and deploy